

# From fine-grained properties to broad principles for gradual argumentation: A principled spectrum

Pietro Baroni<sup>a</sup>, Antonio Rago<sup>b,\*</sup>, Francesca Toni<sup>b</sup>

<sup>a</sup> Dip.to di Ingegneria dell'Informazione, Università degli Studi di Brescia, Italy

<sup>b</sup> Dept. of Computing, Imperial College London, UK

## ARTICLE INFO

### Article history:

Received 18 July 2018

Received in revised form 9 October 2018

Accepted 28 November 2018

Available online 3 December 2018

### Keywords:

Bipolar argumentation

Quantitative argumentation

## ABSTRACT

The study of properties of gradual evaluation methods in argumentation has received increasing attention in recent years, with studies devoted to various classes of frameworks/methods leading to conceptually similar but formally distinct properties in different contexts. In this paper we provide a novel systematic analysis for this research landscape by making three main contributions. First, we identify groups of conceptually related properties in the literature, which can be regarded as based on common patterns and, using these patterns, we evidence that many further novel properties can be considered. Then, we provide a simplifying and unifying perspective for these groups of properties by showing that they are all implied by novel parametric principles of (either strict or non-strict) balance and monotonicity. Finally, we show that (instances of) these principles (and thus the group, literature and novel properties that they imply) are satisfied by several quantitative argumentation formalisms in the literature, thus confirming the principles' general validity and utility to support a compact, yet comprehensive, analysis of properties of gradual argumentation.

© 2018 Elsevier Inc. All rights reserved.

## 1. Introduction

In argument-based reasoning, *Abstract Argumentation Frameworks* (AFs) and their semantics [1] are a well-known formalism to represent at an abstract level the existence of conflicts between arguments, expressed as a binary relation of *attack* amongst *arguments* and to evaluate the acceptance status of the arguments on the basis of the attack relation. While this relatively simple formalism has proven to be extremely useful to study the general properties of conflict management and to be able to capture several concrete formalisms as special cases [1], some of its underlying assumptions can be rather restrictive in some settings. For this reason, in recent years several ways of extending or modifying the traditional AF formalism have been considered. For example, a *support* relation among arguments (and its role in argument evaluation) can be considered alongside (e.g. see [2]) or instead of (e.g. see [3]) the attack relation. While traditional semantics [1] produce simple assessments of argument acceptance or, equivalently, acceptance labellings for arguments [4], finer-grain *gradual evaluation* methods, based on numerical scales or rankings, can be used for AFs (e.g. see [5–8]). These gradual evaluation methods may be complemented by an *initial evaluation* (e.g. called ‘base score’ in [9] and ‘weighting’ in [3]) of each argument, reflecting its assessment before its dialectical relations to other arguments are considered. These base scores may be *fixed*, i.e. identical for all arguments which are considered a-priori equal as in AFs, or *variable*, representing, for example, the authoritativeness

\* Corresponding author.

E-mail addresses: [pietro.baroni@unibs.it](mailto:pietro.baroni@unibs.it) (P. Baroni), [a.rago15@imperial.ac.uk](mailto:a.rago15@imperial.ac.uk) (A. Rago), [ft@imperial.ac.uk](mailto:ft@imperial.ac.uk) (F. Toni).

of their source, the credibility of their premises, the importance of the goals they support or, as in [10], votes they have received.

These interchangeable characteristics have given rise to a wide range of extensions of argumentation frameworks and gradual evaluation methods, some of which are not necessarily dissimilar to one another. Numerous works (see [5,7,11–13,8]) use standard AFs to calculate differing strengths with different semantics. Social AFs (see [10]) and Weighted Argumentation Graphs (WAGs, as defined in [14]) both incorporate a variable base score to AFs but their strengths are calculated on different scales. Variable base scores are also used in Support Argumentation Frameworks (SAFs, see [3]) but here supports are the sole relation, in place of attacks. Both attacks and supports are present with a fixed base score in Bipolar Argumentation Frameworks (BFs, see [2]) and alongside variable base scores in Weighted Bipolar Argumentation Graphs (BAGs, see [15]) and Quantitative Argumentation Debate (QuAD) frameworks (see [9]).

Given the variety of gradual evaluation methods possible, as witnessed by the many proposed in the literature, several works include (e.g. [5,10]) or are completely devoted to (e.g. [13,12]) the definition and study of properties that these methods (should) satisfy and that can be useful for their analysis, classification and design. This has given rise to a relatively large and varied body of properties, which are often somehow similar in spirit and may share basic intuitions but are presented in technically different contexts and are investigated separately. This multiplication of similar but different studies, while potentially fruitful, carries the risk of concealing underlying common roots and possible overlappings.

In this paper we take a step towards the unification of efforts within this research trend, first defining generic *Quantitative Bipolar Argumentation Frameworks* (QBAFs), in which each argument has a (possibly empty) set of attackers, a (possibly empty) set of supporters, and an initial evaluation (possibly the same for all arguments) on a chosen *scale*, all contributing to a final argument evaluation, provided by a *strength* function, on the chosen *scale* (Section 2). Restrictions of these QBAFs are able to represent the various extensions of AFs from the literature, thereby providing a generalised framework to unify the many semantics and properties that we consider. This allows us to reformulate 33 literature properties and show that they are implied by 11 *group properties*, each based on a basic idea which can give rise to many variants, thus highlighting many more fine-grained properties not yet considered in the literature (Section 3). We then show that the 11 group properties are in turn implied by four novel basic *parametric principles*, (strict and non-strict) *balance* and (strict and non-strict) *monotonicity*, which thus have the potential to support a greatly simplified analysis of actual formalisms (Section 4). We then confirm this potential by demonstrating that suitable instances of these principles are satisfied by a variety of existing gradual argumentation formalisms from previous literature (Section 5) before comparing with related works and concluding (Section 6).

This paper is a substantial development of the initial results presented in [16] and in particular extends this previous work as follows: the proposed formalisms have been slightly adjusted for a better presentation, additional properties from recent literature have been included in the analysis of Sections 2 and 3; applications have been suggested for the properties in Section 4; a number of additional approaches have been analysed with respect to the proposed principles in Section 5 (Sections 5.2, 5.3, 5.8, 5.9 and 5.10 describe the newly considered semantics) adding counter-examples for the cases where the approaches do not satisfy the principles; a more extensive discussion has been included in Section 6; all non trivial proofs have been included in Appendix A and a systematic listing of properties implied by our principles while not considered in the previous literature has been provided in Appendix B.

## 2. Preliminaries

Let  $\mathbb{I}$  be a set equipped with a preorder  $\leq$  where, as usual,  $a < b$  denotes  $a \leq b$  and  $b \not\leq a$ .  $\mathbb{I}$  may (but is not required to) contain top ( $\top$ ) and bottom ( $\perp$ ) values, such that if  $\top \in \mathbb{I}$ , then  $i < \top$  for all  $i \in \mathbb{I} \setminus \{\top\}$  and, if  $\perp \in \mathbb{I}$ , then  $\perp < i$  for all  $i \in \mathbb{I} \setminus \{\perp\}$ . For example,  $\mathbb{I} = [0, 1]$  with  $\top = 1$ ,  $\perp = 0$ , or  $\mathbb{I} = \{\text{low}, \text{medium}, \text{high}\}$  with  $\text{low} \leq \text{medium} \leq \text{high}$  and  $\top = \text{high}$ ,  $\perp = \text{low}$ , or  $\mathbb{I} = \{\text{in}, \text{out}, \text{und}\}$  with  $\text{out} \leq \text{und} \leq \text{in}$  and  $\top = \text{in}$ ,  $\perp = \text{out}$ , or  $\mathbb{I} = (0, \infty)$ . A QBAF assigns relations of *attack* and *support* and an initial evaluation (base score) in  $\mathbb{I}$  to arguments.

**Definition 1.** A *Quantitative Bipolar Argumentation Framework* (QBAF) is a quadruple  $\langle \mathcal{X}, \mathcal{R}^-, \mathcal{R}^+, \tau \rangle$  consisting of a finite set  $\mathcal{X}$  of arguments, a binary (*attack*) relation  $\mathcal{R}^-$  on  $\mathcal{X}$ , a binary (*support*) relation  $\mathcal{R}^+$  on  $\mathcal{X}$  and a total function  $\tau : \mathcal{X} \rightarrow \mathbb{I}$ ; for any  $\alpha \in \mathcal{X}$ , we call  $\tau(\alpha)$  the *base score* of  $\alpha$ .

Some comments on Definition 1 are in order.

First, we assume that the set  $\mathcal{X}$  is finite since some properties we consider from the literature rely on this assumption and are based on intuitions which are valid only in the finite case, e.g. that if there is a bijection between two sets one cannot be a strict subset of the other. Considering the infinite case and revising the properties' definitions accordingly is an interesting direction of future work.

Second, while we do not make any assumption on the nature and meaning of the base score, it is worth remarking that this abstract notion corresponds to an assessment of arguments which precedes the consideration of the relations of attack and support with other arguments. This notion admits several concrete counterparts. For instance the base score may be determined on the basis of positive and negative votes received by arguments from an audience in a social evaluation context [10,17] or, in a technical debate context, can be the outcome of an expert assessment [9]. In other approaches the base score arises from an evaluation of the structural components of the argument. In particular in [18] the first “tier” in

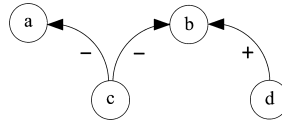


Fig. 1. Example QBAF visualised as a graph.

the evaluation of argument strength is called the *support dimension* and concerns assessing the premises and the rules used in the construction of arguments to define a *degree of support* independently of the presence of other arguments. In general, the study of suitable methods to define arguments' base scores in different application contexts is an active research subject (see for instance the discussion of the notion of *weight of argument* based on imprecise probabilities in [19]) and the notions mentioned above can be combined together, e.g. by putting votes on arguments already equipped with a degree of support, giving rise to articulated mechanisms of base score definition. We leave these aspects to future work, remarking that the abstract notion of base score we use is compatible with any underlying generating mechanism.

QBAFs can be visualised as graphs, e.g. Fig. 1 visualises  $\langle \{a, b, c, d\}, \{(c, a), (c, b)\}, \{(d, b)\}, \tau \rangle$  (for any  $\tau$ ).

In the remainder of the paper, unless specified otherwise, we assume as given a generic QBAF  $\mathcal{Q} = \langle \mathcal{X}, \mathcal{R}^-, \mathcal{R}^+, \tau \rangle$ .

Arguments in a QBAF have a final evaluation (strength):

**Definition 2.** For any  $\alpha \in \mathcal{X}$ , the *strength* of  $\alpha$  is given by  $\sigma(\alpha)$  where  $\sigma : \mathcal{X} \rightarrow \mathbb{I}$  is a total *strength function*. For any set of arguments  $A \subseteq \mathcal{X}$ , we refer to the multiset  $\{\sigma(\beta) | \beta \in A\}$  as  $\sigma(A)$ .<sup>1</sup>

If  $\perp \in \mathbb{I}$ , arguments with  $\perp$  strength may or not play a role in evaluating arguments they attack or support. Thus, we consider two alternative notions of equivalence of sets of arguments as two alternative comparative notions of strength: those which take into account arguments with the  $\perp$  strength and those which do not.

**Definition 3.** Let  $*$  be either  $\sigma \perp$  or  $\sigma \not\perp$ . For  $Z \subseteq \mathcal{X}$ , let  $Z^*$  denote  $\sigma(Z)$  if  $*$  is  $\sigma \perp$  and  $\sigma(Z) \setminus \{\sigma(z) \in \sigma(Z) | \sigma(z) = \perp\}$  if  $*$  is  $\sigma \not\perp$ . Then, for  $A, B \subseteq \mathcal{X}$ ,  $A$  is  *$*$ -strength equivalent* to  $B$ , denoted  $A =_* B$ , iff  $A^* = B^*$ ;  $A$  is *at least as  $*$ -strong as*  $B$ , denoted  $A \geq_* B$ , iff there exists an injective mapping  $f$  from  $B^*$  to  $A^*$  such that  $\forall \alpha \in B^*, \sigma(f(\alpha)) \geq \sigma(\alpha)$ ;  $A$  is  *$*$ -stronger than*  $B$ , denoted  $A >_* B$ , iff  $A \geq_* B$  and  $B \not\geq_* A$ .

It can be noted that if a set is  *$*$ -stronger than* another it exhibits a form of *dominance* over it.

**Lemma 1.** For any  $A, B \subseteq \mathcal{X}$  such that  $A \geq_* B$ ,  $A >_* B$  iff  $|B^*| < |A^*|$  or for every injective mapping  $f$  from  $B^*$  to  $A^*$   $\exists \beta \in B^* : \sigma(f(\beta)) \not\geq \sigma(\beta)$ .

To exemplify, if we take the QBAF in Fig. 1 with some  $\sigma$  such that  $\sigma(c) = \perp$  and  $\sigma(d) = \top$ , then comparing the sets of attackers ( $\{c\}$ ) and supporters ( $\emptyset$ ) of  $a$  we obtain  $\{c\} >_{\sigma \perp} \emptyset$  but  $\{c\} =_{\sigma \not\perp} \emptyset$ , however, for both choices of  $*$  when we compare similarly the attackers ( $\{c\}$ ) and supporters ( $\{d\}$ ) of  $b$ , we get  $\{c\} <_* \{d\}$ .

It is worth noting that Definition 3 for  $*$  is  $\sigma \perp$  is analogous to the notion of *group comparison* in [7,13].

In the remainder of the paper  $*$  is either  $\sigma \perp$  or  $\sigma \not\perp$ .

**Definition 4.** For any  $\alpha \in \mathcal{X}$ , the set of *attackers* of  $\alpha$  is  $\mathcal{R}^-(\alpha) = \{\beta \in \mathcal{X} | (\beta, \alpha) \in \mathcal{R}^-\}$  and the set of *supporters* of  $\alpha$  is  $\mathcal{R}^+(\alpha) = \{\beta \in \mathcal{X} | (\beta, \alpha) \in \mathcal{R}^+\}$ . Also,  $\mathcal{R}_*^-(\alpha) = \mathcal{R}^-(\alpha)$  if  $*$  is  $\sigma \perp$  and  $\mathcal{R}_*^-(\alpha) = \mathcal{R}^-(\alpha) \setminus \{\beta \in \mathcal{R}^-(\alpha) | \sigma(\beta) = \perp\}$  if  $*$  is  $\sigma \not\perp$ . Similarly,  $\mathcal{R}_*^+(\alpha) = \mathcal{R}^+(\alpha)$  if  $*$  is  $\sigma \perp$  and  $\mathcal{R}_*^+(\alpha) = \mathcal{R}^+(\alpha) \setminus \{\beta \in \mathcal{R}^+(\alpha) | \sigma(\beta) = \perp\}$  if  $*$  is  $\sigma \not\perp$ .

The notations  $\mathcal{R}_*^-, \mathcal{R}_*^+$  and  $=_*$  will be needed to capture properties from the literature. We use  $\mathcal{R}_*^-(\alpha)$  ( $\mathcal{R}_*^+(\alpha)$ ) to denote the set of attackers (supporters, resp.) of an argument  $\alpha$  including or excluding, depending on the choice of  $*$ , those with bottom strength. For example, in the QBAF in Fig. 1 with some  $\sigma$  such that  $\sigma(c) = \perp$  and  $\sigma(d) = \top$ ,  $\mathcal{R}_{\sigma \perp}^-(b) = \{c\}$  but  $\mathcal{R}_{\sigma \not\perp}^-(b) = \emptyset$ , whereas for both choices of  $*$ ,  $\mathcal{R}_*^+(b) = \{d\}$ . Note that  $\mathcal{R}_*^-(\alpha) = \mathcal{R}_*^-(\beta)$  implies  $\mathcal{R}^-(\alpha) =_* \mathcal{R}^-(\beta)$  but not vice versa (similarly for supporters). Moreover,  $\mathcal{R}_*^-(\alpha) = \emptyset$  is equivalent to  $\mathcal{R}^-(\alpha) =_* \emptyset$  (similarly for supporters).

The definition of our group properties and principles will also be parametric w.r.t. an operator over  $\mathbb{I}$ ,  $\ll$ , expressing a form of strict comparison between values in  $\mathbb{I}$ . The basic requirement for this operator is that  $< \subseteq \ll \subseteq \leq$  and, naturally,  $m \gg n$  iff  $n \ll m$ . Throughout the paper, we will consider the following instances of  $\ll$ , in addition to  $\ll = <$  and  $\ll = \leq$ :

$\ll = <_\times$ , where  $m <_\times n$  iff  $m < n$  or  $m = n = \times$ , where  $\times$  is one of  $\perp, \top$ , or some other specified element of  $\mathbb{I}$ .

<sup>1</sup> In this paper we use the same notation for multisets as for sets.

**Table 1**

Choices of QBAFs, base score (where any  $i \in \mathbb{I}$  represents  $\tau(\alpha) = i \ \forall \alpha \in \mathcal{X}$  and  $-$  represents any variable base score) and  $\mathbb{I}$  for the literature considered.

| Literature                       | QBAF  | $\tau$ | $\mathbb{I}$                                         |
|----------------------------------|-------|--------|------------------------------------------------------|
| Matt & Toni, 2008 [5]            | aQBAF | 1      | [0, 1]                                               |
| Amgoud & Ben-Naim, 2013 [7]      | aQBAF | T      | ranking                                              |
| Thimm & Kern-Isberner, 2014 [11] | aQBAF | T      | ranking                                              |
| Amgoud & Ben-Naim, 2016 [12]     | aQBAF | 1      | [0, 1]                                               |
| Bonzon et al., 2016 [13]         | aQBAF | T      | pre-order                                            |
| Amgoud et al., 2016 [8]          | aQBAF | T      | [1, $\infty$ )                                       |
| Leite & Martins, 2011 [10]       | aQBAF | -      | $\mathbb{I} \supseteq \{\perp, \top\}$ (ordered set) |
| Amgoud et al., 2017 [14]         | aQBAF | -      | [0, 1]                                               |
| Amgoud & Ben-Naim, 2016 [3]      | sQBAF | -      | [0, 1]                                               |
| Amgoud et al., 2008 [2]          | QBAF  | 0      | [-1, 1]                                              |
| Amgoud & Ben-Naim, 2017 [15]     | QBAF  | -      | [0, 1]                                               |
| Baroni et al., 2015 [9]          | QBAF  | -      | [0, 1]                                               |
| Rago et al., 2016 [20]           | QBAF  | -      | [0, 1]                                               |

### 3. Grouping literature properties

In this section we show that 33 *literature properties* (Ps) introduced and analysed separately for one gradual argumentation formalism or another,<sup>2</sup> can be seen as instances of or are directly implied by 11 *group properties* (GPs) that we introduce for *generic QBAFs* in terms of parameters  $\mathbb{I}$ ,  $*$ ,  $\ll$ . In order to describe the Ps in terms of QBAFs, we introduce a number of restrictions of QBAFs which correspond to several existing formalisms in the literature. Let a QBAF  $\mathcal{Q}$  be referred to as:

$$\begin{aligned} \text{aQBAF} & \quad \text{if } \forall \alpha \in \mathcal{X}, \mathcal{R}^+(\alpha) = \emptyset \\ \text{sQBAF} & \quad \text{if } \forall \alpha \in \mathcal{X}, \mathcal{R}^-(\alpha) = \emptyset \end{aligned}$$

We also introduce here the notion of QBAFs having *fixed* base scores for their arguments, i.e.  $\forall \alpha \in \mathcal{X}, \tau(\alpha) = i$  for some  $i \in \mathbb{I}$ . Some properties and semantics in the literature have been defined for (restrictions of) QBAFs with fixed base scores, which are generalised by our properties (defined for unrestricted QBAFs).

Then, for the purposes of our analysis in this paper, aQBAFs (i.e. QBAFs with empty support) with fixed base score T for all arguments correspond to AFs [1] (formally, a generic AF  $\langle \mathcal{X}, \mathcal{R}^- \rangle$  can be understood as a QBAF  $\langle \mathcal{X}, \mathcal{R}^-, \emptyset, \tau \rangle$  where  $\forall \alpha \in \mathcal{X}, \tau(\alpha) = i \in \mathbb{I}$  for some  $i$ ), aQBAFs correspond to Social AFs [10] (where base scores are determined by votes) and to Weighted Argumentation Graphs [14], sQBAFs (i.e. QBAFs with empty attack) correspond to Support Argumentation Frameworks [3], and QBAFs with a fixed base score for all arguments correspond to Bipolar Argumentation Frameworks, as in [2]. Quantitative Argumentation Debate (QuAD) frameworks [9] and Weighted Bipolar Argumentation Graphs (BAGs) [15] can be seen as generic QBAFs for the specific choice of  $\mathbb{I} = [0, 1]$ .<sup>3</sup>

Table 1 summarises the concrete choices of base score  $\tau$ ,<sup>4</sup> and the appropriate special type of QBAF for the approaches from the literature recalled in this paper and hence for the Ps defined therein (we will give the other choices of parameters in Table 2 later). Note that if a specific value  $i \in \mathbb{I}$  is given for  $\tau$  in the table, then  $\tau(\alpha) = i \ \forall \alpha \in \mathcal{X}$ , whereas a  $-$  indicates that  $\tau$  is any generic base score as in Definition 1.

Our analysis in this section results in a principled catalogue of Ps, collated into groups, and, as a relevant side-effect, the (implicit) definition of several other properties, variants of Ps, which, to the best of our knowledge, have not been considered in the literature before their existence was evidenced in [16]. To give a simple illustrative example, consider the property of *maximality*, stating for attack-only frameworks that an argument with no attackers has strength equal to its base score [14], and the property of *minimality*, stating for support-only frameworks that an argument with no supporters has strength equal to its base score [3]. Clearly, they share the same basic idea that if an argument is not the dialectical target of any arguments, then its strength coincides with its base score as it cannot be, positively or negatively, affected by anything. Thus these two Ps can be grouped (intuitively by conjoining them) giving a further property for QBAFs, namely *stability* [15]. Further, the basic idea of this GP admits variants concerning whether arguments with bottom strength should count as effective attackers/supporters or not, i.e. whether the sets whose emptiness has to be checked are those of attackers/supporters *tout court* or with non-bottom strength. Each variant of the basic idea is a reasonable property in its own right but not all variants have been given in the literature, in particular the case where bottom strength arguments do not count has not been introduced explicitly for aQBAF and sQBAF.

<sup>2</sup> Some properties were introduced referring to a generic notion of strength, others were proven for specific strength proposals.

<sup>3</sup> See Section 5 for a formal description of the choices of parameters and restrictions used in several evaluation methods in the literature.

<sup>4</sup> The ranges in Table 1 can be identified by inspection of the papers; rankings/orders can be naturally mapped onto  $\mathbb{I}$ .

**Table 2**

GPs and their instances (in bold) or implied instances (in normal font) in the literature. Note that GP1, GP4–GP6 do not use  $\ll$ . GP2, GP4, GP7, GP10 have no sQBAF variants and GP3, GP5, GP8, GP11 have no aQBAF variants.

|      |       | $\sigma \perp$ |                 |                 |            | $\sigma \perp$ |                 |            |        |
|------|-------|----------------|-----------------|-----------------|------------|----------------|-----------------|------------|--------|
|      |       | $<$            | $<\perp$        | $<\top$         | $\leq$     | $<$            | $<\perp$        | $<\top$    | $\leq$ |
| GP1  | aQBAF |                |                 | <b>P1</b>       |            |                |                 | –          |        |
|      | sQBAF |                |                 | <b>P2</b>       |            |                |                 | –          |        |
|      | QBAF  |                |                 | <b>P3</b>       |            |                |                 | <b>P4</b>  |        |
| GP2  | aQBAF | –              | –               | –               | –          | –              | <b>P5</b>       | –          | –      |
|      | QBAF  | –              | –               | –               | –          | –              | –               | –          | –      |
| GP3  | sQBAF | –              | –               | –               | –          | –              | –               | <b>P6</b>  | –      |
|      | QBAF  | –              | –               | –               | –          | –              | –               | –          | –      |
| GP4  | aQBAF |                |                 | –               |            |                |                 | <b>P7</b>  |        |
|      | QBAF  |                |                 | –               |            |                |                 | –          |        |
| GP5  | sQBAF |                |                 | –               |            |                |                 | <b>P8</b>  |        |
|      | QBAF  |                |                 | –               |            |                |                 | –          |        |
| GP6  | aQBAF |                |                 | <b>P9, P15</b>  |            |                |                 | <b>P10</b> |        |
|      | sQBAF |                |                 | <b>P11</b>      |            |                |                 | <b>P12</b> |        |
|      | QBAF  |                |                 | <b>P13</b>      |            |                |                 | <b>P14</b> |        |
| GP7  | aQBAF | <b>P17</b>     | –               | –               | <b>P18</b> | –              | <b>P19, P20</b> | –          | –      |
|      | QBAF  | –              | –               | –               | <b>P16</b> | –              | –               | –          | –      |
| GP8  | sQBAF | –              | –               | –               | –          | –              | –               | <b>P22</b> | –      |
|      | QBAF  | –              | –               | –               | <b>P21</b> | –              | –               | –          | –      |
| GP9  | aQBAF | –              | <b>P23</b>      | –               | –          | –              | –               | –          | –      |
|      | sQBAF | –              | –               | <b>P24</b>      | –          | –              | –               | –          | –      |
|      | QBAF  | –              | –               | –               | –          | –              | –               | –          | –      |
| GP10 | aQBAF | <b>P25</b>     | <b>P26, P27</b> | –               | –          | –              | –               | –          | –      |
|      | QBAF  | –              | –               | –               | –          | –              | –               | –          | –      |
| GP11 | sQBAF | –              | –               | <b>P28, P29</b> | –          | –              | –               | –          | –      |
|      | QBAF  | –              | –               | –               | –          | –              | –               | –          | –      |

In this section we give formally our GPs in turn, each preceded by a short informal description of the underlying basic idea and followed by formal results on Ps<sup>5</sup> that are instances of or directly implied by the GP.<sup>6</sup> For ease of reference and to disambiguate some homonymies, we number Ps. Tables 1 and 2 summarise the choices of parameters for capturing Ps as (implied by) instances of our GPs. Table 2 also indicates with hyphens instances of GPs without an explicit counterpart in the literature. Thus, each hyphen corresponds to (at least) one new property, instance of one of our GPs. As an illustration, the first row of Table 2 specifies that P1 (i.e. *maximality*, discussed earlier) is the instance of GP1 (defined later) for aQBAFs with  $\ast = \sigma \perp$ , but no corresponding instance of GP1 for aQBAFs with  $\ast = \sigma \perp$  has been considered. In Appendix B, we will give explicitly the properties corresponding to these hyphens, interesting in their own right. These instantiations of the GPs turn out to be appropriate in several different potential application contexts (see [21]) but this is outside the scope of this work. In the remainder, unless specified otherwise,  $\alpha, \beta \in \mathcal{X}$  in the given generic QBAF  $\mathcal{Q} = \langle \mathcal{X}, \mathcal{R}^-, \mathcal{R}^+, \tau \rangle$ .

**Basic Idea 1.** *The strength of an argument differs from its base score only if the argument is the dialectical target of other arguments, i.e. the strength is equal to the base score if the argument is not the dialectical target of other arguments.*

**GP1.** If  $\mathcal{R}_*^-(\alpha) = \emptyset$  and  $\mathcal{R}_*^+(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ .

**Proposition 1.** *The following are instances of GP1<sup>7</sup>:*

<sup>5</sup> The trivial (i.e. obtainable by inspection of the definitions) proofs for Propositions 1 to 15, 18, 20 and 21 have been omitted. The (non-trivial) proofs for the other Propositions are included in Appendix A.

<sup>6</sup> We remark that some GPs could be formulated in different ways which are conceptually similar but formally slightly different. We chose to use definitions that are formally as close as possible to the existing properties in the literature.

<sup>7</sup> Each P is reformulated using our notation, but this reformulation is equivalent to the original formulation in the literature. Here and throughout, instantiation amounts to the choices of QBAF and  $\mathbb{I}$  in Table 1 and of  $\ast$  and  $\ll$ , if applicable, in Table 2. By showing that Ps are instances of GPs we show that the former are equivalent to (suitable instances of) the latter.

- P1. Maximality** [12,14], [10, Prop. 11] – For any  $\alpha \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ .  
**P2. Minimality** [3] – For any  $\alpha \in \mathcal{X}$ , if  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ .  
**P3. Stability** [15] – For any  $\alpha \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ .  
**P4. Equation (4)** [9] – For any  $\alpha \in \mathcal{X}$ , if  $\nexists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\nexists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\sigma(\alpha) = \tau(\alpha)$ .

Note the difference between the way in which GP1 for QBAFs with  $* = \sigma \perp$  is expressed and that for P4, despite the two properties being equivalent.

The hyphens in Table 2 in the rows for GP1 correspond to the following novel properties:

GP1 for  $\sigma \perp$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\nexists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\sigma(\alpha) = \tau(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\nexists \beta \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\sigma(\alpha) = \tau(\alpha)$ .

We give the properties corresponding to the other hyphens from Table 2 in Appendix B.

**Basic Idea 2.** In the absence of supporters, if there is at least an attacker then the strength of an argument is lower than its base score.

**GP2.** If  $\mathcal{R}_*^-(\alpha) \neq \emptyset$  and  $\mathcal{R}_*^+(\alpha) = \emptyset$  then  $\sigma(\alpha) \ll \tau(\alpha)$ .

**Proposition 2.** GP2 implies the following:

**P5. Weakening** [12,14] – For any  $\alpha \in \mathcal{X}$ , if  $\tau(\alpha) > \perp$  and  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\sigma(\alpha) < \tau(\alpha)$ .

**Basic Idea 3.** In the absence of attackers, if there is at least a supporter then the strength of an argument is greater than its base score.

**GP3.** If  $\mathcal{R}_*^-(\alpha) = \emptyset$  and  $\mathcal{R}_*^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) \ll \sigma(\alpha)$ .

**Proposition 3.** GP3 implies the following:

**P6. Strengthening** [3] – For any  $\alpha \in \mathcal{X}$ , if  $\tau(\alpha) < \top$  and  $\exists \beta \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\tau(\alpha) < \sigma(\alpha)$ .

**Basic Idea 4.** If the strength of an argument is lower than its base score then the argument has at least one attacker.

**GP4.** If  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}_*^-(\alpha) \neq \emptyset$ .

**Proposition 4.** The following is an instance of GP4:

**P7. Weakening Soundness** [12,14] – For any  $\alpha \in \mathcal{X}$  with  $\tau(\alpha) > \perp$ , if  $\sigma(\alpha) < \tau(\alpha)$  then  $\exists \beta \in \mathcal{R}^-(\alpha)$  with  $\sigma(\beta) > \perp$ .

**Basic Idea 5.** If the strength of an argument is higher than its base score then the argument has at least one supporter.

**GP5.** If  $\sigma(\alpha) > \tau(\alpha)$  then  $\mathcal{R}_*^+(\alpha) \neq \emptyset$ .

**Proposition 5.** The following is an instance of GP5:

**P8. Strengthening Soundness** [3] – For any  $\alpha \in \mathcal{X}$ , if  $\sigma(\alpha) > \tau(\alpha)$  then  $\exists \beta \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\beta) > \perp$ .

**Basic Idea 6.** Arguments with equal conditions in terms of attackers, supporters and base score have the same strength.

**GP6.** If  $\mathcal{R}^-(\alpha) =_* \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) =_* \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$ .

**Proposition 6.** The following are instances of GP6:

- P9. Equivalence** [12,14] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and there exists a bijective function  $f$  from  $\mathcal{R}^-(\alpha)$  to  $\mathcal{R}^-(\beta)$  such that  $\forall \gamma \in \mathcal{R}^-(\alpha), \sigma(\gamma) = \sigma(f(\gamma))$  then  $\sigma(\alpha) = \sigma(\beta)$ .  
**P10. Neutrality** [12,14] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$ ,  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta) \setminus \{\gamma\}$ ,  $\gamma \in \mathcal{R}^-(\beta)$  and  $\sigma(\gamma) = \perp$  then  $\sigma(\beta) = \sigma(\alpha)$ .  
**P11. Equivalence** [3] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and there exists a bijective function  $f$  from  $\mathcal{R}^+(\alpha)$  to  $\mathcal{R}^+(\beta)$  such that  $\forall \gamma \in \mathcal{R}^+(\alpha), \sigma(\gamma) = \sigma(f(\gamma))$  then  $\sigma(\alpha) = \sigma(\beta)$ .  
**P12. Dummy** [3] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$ ,  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta) \setminus \{\gamma\}$  and  $\gamma \in \mathcal{R}^+(\beta)$  with  $\sigma(\gamma) = \perp$  then  $\sigma(\beta) = \sigma(\alpha)$ .



- P13. Bi-variate Equivalence** [15] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and there exists a bijective function  $f$  from  $\mathcal{R}^-(\alpha)$  to  $\mathcal{R}^-(\beta)$  such that  $\forall \gamma \in \mathcal{R}^-(\alpha), \sigma(\gamma) = \sigma(f(\gamma))$  and there exists a bijective function  $f'$  from  $\mathcal{R}^+(\alpha)$  to  $\mathcal{R}^+(\beta)$  such that  $\forall \delta \in \mathcal{R}^+(\alpha), \sigma(\delta) = \sigma(f'(\delta))$  then  $\sigma(\alpha) = \sigma(\beta)$ .
- P14. Neutrality** [15] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta), \mathcal{R}^-(\alpha) \subseteq \mathcal{R}^-(\beta), \mathcal{R}^+(\alpha) \subseteq \mathcal{R}^+(\beta), \mathcal{R}^-(\beta) \cup \mathcal{R}^+(\beta) = \mathcal{R}^-(\alpha) \cup \mathcal{R}^+(\alpha) \cup \{\gamma\}$  and  $\sigma(\gamma) = \perp$ , then  $\sigma(\beta) = \sigma(\alpha)$ .

**Proposition 7.** GP6 implies the following:

- P15. Non-Attacked Equivalence** [13] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^-(\beta) = \emptyset$  then  $\sigma(\alpha) = \sigma(\beta)$ .

**Basic Idea 7.** A strictly larger<sup>8</sup> set of attackers determines a lower strength.

- GP7.** If  $\mathcal{R}_*^-(\alpha) \subsetneq \mathcal{R}_*^-(\beta)$  and  $\mathcal{R}_*^+(\alpha) = \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \ll \sigma(\alpha)$ .

**Proposition 8.** The following is an instance of GP7:

- P16. Equation (2)** [9], [20, Prop. 12] – For any  $\alpha, \beta \in \mathcal{X}$ , such that  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$ , if  $\mathcal{R}^-(\alpha) \subsetneq \mathcal{R}^-(\beta)$  then  $\sigma(\alpha) \geq \sigma(\beta)$ .

**Proposition 9.** GP7 implies the following:

- P17. Void Precedence** [7,13], [8, Th. 5] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^-(\beta) \neq \emptyset$  then  $\sigma(\alpha) > \sigma(\beta)$ .
- P18. Weak Void Precedence** [11] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$  then  $\sigma(\alpha) \geq \sigma(\beta)$ .
- P19. Triggering** [12] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\sigma(\alpha) > \perp, \forall \gamma \in \mathcal{R}^-(\alpha), \sigma(\gamma) = \perp$ , and  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta) \setminus \{\delta\}$  for some  $\delta \in \mathcal{R}^-(\beta)$  with  $\sigma(\delta) > \perp$  then  $\sigma(\beta) < \sigma(\alpha)$ .
- P20. Counting** [14]<sup>9</sup> – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta), \sigma(\alpha) > \perp$  and  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta) \setminus \{\gamma\}$  for some  $\gamma \in \mathcal{R}^-(\beta)$  with  $\sigma(\gamma) > \perp$  then  $\sigma(\beta) < \sigma(\alpha)$ .

**Basic Idea 8.** A strictly larger set of supporters determines a higher strength.

- GP8.** If  $\mathcal{R}_*^-(\alpha) = \mathcal{R}_*^-(\beta)$  and  $\mathcal{R}_*^+(\alpha) \subsetneq \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \ll \sigma(\beta)$ .

**Proposition 10.** The following is an instance of GP8:

- P21. Equation (3)** [9], [20, Prop. 13] – For any  $\alpha, \beta \in \mathcal{X}$  such that  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ , if  $\mathcal{R}^+(\alpha) \subsetneq \mathcal{R}^+(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .

**Proposition 11.** GP8 implies the following:

- P22. Counting** [3] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta), \mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta) \setminus \{\gamma\}$  for some  $\gamma \in \mathcal{R}^+(\beta)$  with  $\sigma(\gamma) > \perp$  and  $\sigma(\alpha) < \top$ , then  $\sigma(\beta) > \sigma(\alpha)$ .

**Basic Idea 9.** A higher base score gives a higher strength.

- GP9.** If  $\mathcal{R}_*^-(\alpha) = \mathcal{R}_*^-(\beta), \mathcal{R}_*^+(\alpha) = \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) \ll \sigma(\beta)$ .

**Proposition 12.** GP9 implies the following:

- P23. Proportionality** [14] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta), \tau(\beta) > \tau(\alpha)$  and either  $\sigma(\alpha) > \perp$  or  $\sigma(\beta) > \perp$ , then  $\sigma(\beta) > \sigma(\alpha)$ .
- P24. Coherence** [3] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta), \tau(\beta) > \tau(\alpha)$  and  $\sigma(\alpha) < \top$ , then  $\sigma(\beta) > \sigma(\alpha)$ .

**Basic Idea 10.** A weaker set of attackers determines a higher strength.

- GP10.** If  $\mathcal{R}^-(\alpha) <_* \mathcal{R}^-(\beta), \mathcal{R}_*^+(\alpha) = \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \ll \sigma(\alpha)$ .

<sup>8</sup> Note that this and several other GPs in the remainder admit equivalent alternative formulations by “switching” the adjectives, like “A strictly smaller set of attackers determines a higher strength”.

<sup>9</sup> We omit here the more restrictive form of Counting given in the earlier [12].

**Table 3**

Some possible combinations of GPs and implied instances in the literature.

|            |       | $\sigma \perp$ |           |          |        | $\sigma \perp$ |           |          |        |
|------------|-------|----------------|-----------|----------|--------|----------------|-----------|----------|--------|
|            |       | $<$            | $< \perp$ | $< \top$ | $\leq$ | $<$            | $< \perp$ | $< \top$ | $\leq$ |
| GP6 & GP7  | aQBAF |                |           | P30      |        | –              | –         | –        | –      |
|            | QBAF  | –              | –         | –        | –      | –              | –         | –        | –      |
| GP6 & GP8  | sQBAF |                |           | P31      |        | –              | –         | –        | –      |
|            | QBAF  | –              | –         | –        | –      | –              | –         | –        | –      |
| GP6 & GP10 | aQBAF |                |           | P32, P33 |        | –              | –         | –        | –      |
|            | QBAF  | –              | –         | –        | –      | –              | –         | –        | –      |

**Proposition 13.** The following is an instance of GP10:

**P25.** Strict Counter-Transitivity [7,13], [8, Th. 8] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \mathcal{R}^-(\beta)$ , then  $\sigma(\alpha) > \sigma(\beta)$ .

**Proposition 14.** GP10 implies the following:

**P26.** Boundedness [12] – For any  $\alpha, \beta \in \mathcal{X}$  with  $\gamma \in \mathcal{R}^-(\alpha)$  and  $\delta \in \mathcal{R}^-(\beta)$  where  $\sigma(\delta) > \sigma(\gamma)$ , if  $\mathcal{R}^-(\beta) \setminus \{\delta\} = \mathcal{R}^-(\alpha) \setminus \{\gamma\}$  and  $\sigma(\alpha) = \perp$  then  $\sigma(\beta) = \perp$ .

**P27.** Reinforcement [14]<sup>10</sup> – For any  $\alpha, \beta \in \mathcal{X}$ , with  $\tau(\alpha) = \tau(\beta)$ , either  $\sigma(\alpha) > \perp$  or  $\sigma(\beta) > \perp$ ,  $\gamma \in \mathcal{R}^-(\alpha)$  and  $\delta \in \mathcal{R}^-(\beta)$  where  $\sigma(\delta) > \sigma(\gamma)$ , if  $\mathcal{R}^-(\beta) \setminus \{\delta\} = \mathcal{R}^-(\alpha) \setminus \{\gamma\}$  then  $\sigma(\beta) < \sigma(\alpha)$ .

**Basic Idea 11.** A stronger set of supporters determines a higher strength.

**GP11.** If  $\mathcal{R}_*^-(\alpha) = \mathcal{R}_*^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_* \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \ll \sigma(\alpha)$ .

**Proposition 15.** GP11 implies the following:

**P28.** Boundedness [3] – For any  $\alpha, \beta \in \mathcal{X}$  with  $\gamma \in \mathcal{R}^+(\alpha)$ ,  $\delta \in \mathcal{R}^+(\beta)$  and  $\sigma(\delta) > \sigma(\gamma)$ , if  $\tau(\alpha) = \tau(\beta)$ ,  $\mathcal{R}^+(\beta) \setminus \{\delta\} = \mathcal{R}^+(\alpha) \setminus \{\gamma\}$  and  $\sigma(\alpha) = \top$  then  $\sigma(\beta) = \top$ .

**P29.** Reinforcement [3] – For any  $\alpha, \beta \in \mathcal{X}$ , with  $\gamma \in \mathcal{R}^+(\alpha)$  and  $\delta \in \mathcal{R}^+(\beta)$  where  $\sigma(\delta) > \sigma(\gamma) > \perp$ , if  $\tau(\alpha) = \tau(\beta)$ ,  $\mathcal{R}^+(\beta) \setminus \{\delta\} = \mathcal{R}^+(\alpha) \setminus \{\gamma\}$  and  $\sigma(\alpha) < \top$  then  $\sigma(\beta) > \sigma(\alpha)$ .

Other Ps are implied by combinations of GPs, as illustrated by the following proposition:

**Proposition 16.**

- GP6 and GP7 with  $* = \sigma \perp$  and with any  $\ll$  imply:

**P30.** Monotony [12], [5, Prop. 6] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^-(\alpha) \subseteq \mathcal{R}^-(\beta)$  then  $\sigma(\alpha) \geq \sigma(\beta)$ .

- GP6 and GP8 with  $* = \sigma \perp$  and with any  $\ll$  imply:

**P31.** Monotony [3] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^+(\alpha) \subseteq \mathcal{R}^+(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .

- GP6 and GP10 with  $* = \sigma \perp$  and with any  $\ll$  imply:

**P32.** Theorem 7 in [8] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and there exists an injective mapping  $f$  from  $\mathcal{R}^-(\alpha)$  to  $\mathcal{R}^-(\beta)$  such that  $\forall \gamma \in \mathcal{R}^-(\alpha)$ ,  $\sigma(f(\gamma)) \geq \sigma(\gamma)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .

**P33.** Counter-Transitivity [7,13] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^-(\alpha) \leq_{\sigma \perp} \mathcal{R}^-(\beta)$ , then  $\sigma(\alpha) \geq \sigma(\beta)$ .

Table 3 shows the possible combinations of GPs and their implied Ps, while highlighting yet more “missing” properties. Note here that although the four Ps are implied by the combinations of properties for any choice of  $\ll$ , the properties themselves are distinct from one another. We give some of the properties corresponding to the hyphens from Table 3 in Appendix B.

#### 4. (Strict) balance/monotonicity principles

GPs capitalise on similarities across argumentation frameworks and instantiation of parameters to synthesise various fine-grained Ps, while pointing to several new ones, but are still numerous. In this section we define more synthetic principles of (strict) balance (Section 4.1) and monotonicity (Section 4.2), in terms of the same parameters as for GPs, and show

<sup>10</sup> We omit here the more restrictive form of Reinforcement given in the earlier [12].



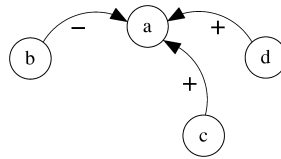


Fig. 2. Example QBAF for strict balance.

that they imply all the GPs (by showing that if the implied GPs' premises hold, then so do the implying principle's, and that the principle's conclusions imply the conclusions of the GPs). As a by-product, any Ps equivalent to or implied by the GPs, as shown in Section 3, are also implied by these principles. Moreover, it turns out that some additional literature properties not implied by GPs are also implied by our principles.

#### 4.1. (Strict) balance

The intuition behind our first principle is that a difference between strength and base score of an argument must correspond to some imbalance between the strengths of its attackers and supporters, for example, from a dialectical viewpoint, because the reasons against an argument are stronger than the reasons for it. Thus, an argument with attackers  $*$ -strength equivalent to its supporters has a strength equal to its base score, while an argument with attackers  $*$ -stronger than ( $*$ -weaker than) its supporters has a strength not greater (not less, resp.) than its base score and cannot attain the top (bottom, resp.) value. (Strict) Balance, admitting multiple instances for different choices of  $*$  and  $\ll$ , expresses this:

**Principle 1.** A strength function  $\sigma$  is:

- *balanced* iff for any  $\alpha \in \mathcal{X}$ :
  1. If  $\mathcal{R}^-(\alpha) =_* \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) = \tau(\alpha)$ .
  2. If  $\mathcal{R}^-(\alpha) >_* \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) \ll \tau(\alpha)$ .
  3. If  $\mathcal{R}^-(\alpha) <_* \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) \gg \tau(\alpha)$ .
- *strictly balanced* iff  $\sigma$  is balanced and for any  $\alpha \in \mathcal{X}$ :
  4. If  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_* \mathcal{R}^+(\alpha)$ .
  5. If  $\sigma(\alpha) > \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) <_* \mathcal{R}^+(\alpha)$ .

For illustration, consider the QBAF given in Fig. 1 with  $\tau(a) = \tau(b)$ ,  $\sigma(c) = \perp$  and  $\sigma(d) > \perp$ . Let  $\sigma$  be balanced and  $\ll = <$ . If  $*$  =  $\sigma \perp$ , by Definition 3,  $\mathcal{R}^-(a) >_* \mathcal{R}^+(a)$  and so, by Point 2 of Principle 1,  $\sigma(a) \ll \tau(a)$ , meaning  $\sigma(a) < \tau(a)$ . However, if  $*$  =  $\sigma \perp$ , by Definition 3,  $\mathcal{R}^-(a) =_* \mathcal{R}^+(a)$  and so, by Point 1 of Principle 1,  $\sigma(a) = \tau(a)$ . Let instead  $\ll = <_{\top}$ . Then, for both choices of  $*$ ,  $\mathcal{R}^-(b) <_* \mathcal{R}^+(b)$  and so, by Point 3 of Principle 1,  $\sigma(b) \gg \tau(b)$ , meaning  $\sigma(b) > \tau(b)$  or  $b$ 's strength is saturated, i.e.  $\sigma(b) = \tau(b) = \top$ .

We foresee different applications requiring strict balance, balance or neither of these principles. Strict balance seems to be a legitimate choice for applications where sets of arguments are evaluated based on a dominance-based set comparison measure such as ours. For example, consider the QBAF shown in Fig. 2 in which  $\mathcal{X} = \{a, b, c, d\}$  with  $\mathcal{R}^-(a) = \{b\}$  and  $\mathcal{R}^+(a) = \{c, d\}$ . If we were looking to analyse a general debate between a group of people in which the strengths of the arguments could be a few discrete values, e.g.  $\{\text{very weak}, \text{weak}, \text{strong}, \text{very strong}\}$ , then the evaluation of arguments may be based on a dominance-based ranking, such as  $\sigma \perp$ , and therefore, for example, it may be desirable that  $\sigma(a) > \tau(a)$  corresponds to  $\{b\} <_{\sigma \perp} \{c, d\}$ . This may not be the case in more gradual applications, for example e-Polling (see [17]), in which the base scores may correspond to the number of votes a particular argument received and so the base scores and strengths are on a continuous scale. Requiring a difference in base score and strength to correspond to our version of set comparison may be restricting in this setting, for example, if  $\sigma(b) = 0.9$ ,  $\sigma(c) = 0.8$  and  $\sigma(d) = 0.8$ , it may be desirable that  $\sigma(a) > \tau(a)$  though  $\{b\} <_{\sigma \perp} \{c, d\}$  does not hold. Finally, some applications may require that balance is not fulfilled. These may include adversarial settings in which arguments' strengths may be based on sets of arguments (see [22]). For example, in the law domain, attacks which are not defended against may present a loss in credibility. In the example this could be shown by the undefended attack from  $b$  reducing  $\sigma(a)$  to  $\perp$ , regardless of  $\sigma(c)$  or  $\sigma(d)$ .

**Proposition 17.** If  $\sigma$  is balanced for some  $*$ ,  $\ll$  then it satisfies GP1 to GP3 for the same  $*$ ,  $\ll$ ; if  $\sigma$  is strictly balanced for some  $*$ ,  $\ll$  then it satisfies GP1 to GP5 for the same  $*$ ,  $\ll$ .

Note that the reverse of Proposition 17 does not hold. For example, if  $a$  is an argument in a QBAF such that  $\mathcal{R}_*^-(a) = \mathcal{R}_*^+(a)$  but  $\sigma(a) \neq \tau(a)$  then GP1–GP5 may hold while  $\sigma$  is not (strictly) balanced.<sup>11</sup>

Note also that other properties have been considered in the literature which are directly implied by balance, as shown by the following proposition.

<sup>11</sup> This is the case for the QuAD algorithm, where  $\mathcal{R}_*^-(a) = \mathcal{R}_*^+(a) \neq \emptyset$ , described later.

**Proposition 18.** *Balance implies the following:*

*Weakening [15] – with  $* = \sigma \perp$  and  $\ll = <_{\perp}$  – For any  $\alpha \in \mathcal{X}$ , if  $\tau(\alpha) > \perp$  and there exists an injective function  $f$  from  $\mathcal{R}^+(\alpha)$  to  $\mathcal{R}^-(\alpha)$  such that  $\forall \beta \in \mathcal{R}^+(\alpha)$ ,  $\sigma(\beta) \leq \sigma(f(\beta))$  and either  $\mathcal{R}_{\sigma \perp}^-(\alpha) \setminus \{f(\gamma) | \gamma \in \mathcal{R}^+\} \neq \emptyset$  or  $\exists \delta \in \mathcal{R}^+(\alpha) : \sigma(\delta) < \sigma(f(\delta))$ , then  $\sigma(\alpha) < \tau(\alpha)$ .*

*Strengthening [15] – with  $* = \sigma \perp$  and  $\ll = <_{\top}$  – For any  $\alpha \in \mathcal{X}$ , if  $\tau(\alpha) < \top$  and there exists an injective function  $f$  from  $\mathcal{R}^-(\alpha)$  to  $\mathcal{R}^+(\alpha)$  such that  $\forall \beta \in \mathcal{R}^-(\alpha)$ ,  $\sigma(\beta) \leq \sigma(f(\beta))$  and either  $\mathcal{R}_{\sigma \perp}^+(\alpha) \setminus \{f(\gamma) | \gamma \in \mathcal{R}^-\} \neq \emptyset$  or  $\exists \delta \in \mathcal{R}^-(\alpha) : \sigma(\delta) < \sigma(f(\delta))$ , then  $\sigma(\alpha) > \tau(\alpha)$ .*

#### 4.2. (Strict) monotonicity

Our second principle requires that the strength of an argument depends monotonically on its base score and on the strengths of its attackers and supporters. From a dialectical viewpoint, the strength of an argument depends exclusively on its intrinsic strength, the reasons for it and the reasons against it, and any strengthening or weakening of these will strengthen or weaken the argument. To define this principle formally, we first define the notion of *shaping triple* of an argument, as follows.

**Definition 5.** For any  $\alpha \in \mathcal{X}$ , the *shaping triple* of  $\alpha$  is  $(\tau(\alpha), \mathcal{R}^+(\alpha), \mathcal{R}^-(\alpha))$ , denoted  $ST(\alpha)$ .

We define partial orders over shaping triples, parametric w.r.t.  $*$ , based on the ordering of their elements. Intuitively, the orders rank the ability of the triples to boost arguments.

**Definition 6.** Given  $\alpha, \beta \in \mathcal{X}$ ,  $ST(\beta)$  is said to be:

- as  $*$ -boosting as  $ST(\alpha)$ , denoted  $ST(\alpha) \simeq_* ST(\beta)$ , iff  $\tau(\alpha) = \tau(\beta)$ ,  $\mathcal{R}^+(\alpha) =_* \mathcal{R}^+(\beta)$ , and  $\mathcal{R}^-(\beta) =_* \mathcal{R}^-(\alpha)$ ;
- at least as  $*$ -boosting as  $ST(\alpha)$ , denoted  $ST(\alpha) \leq_* ST(\beta)$ , iff  $\tau(\alpha) \leq \tau(\beta)$ ,  $\mathcal{R}^+(\alpha) \leq_* \mathcal{R}^+(\beta)$ , and  $\mathcal{R}^-(\beta) \leq_* \mathcal{R}^-(\alpha)$ .
- strictly more  $*$ -boosting than  $ST(\alpha)$ , denoted  $ST(\alpha) <_* ST(\beta)$ , iff  $ST(\alpha) \leq_* ST(\beta)$  and  $ST(\beta) \not\leq_* ST(\alpha)$ .

(Strict) Monotonicity is defined by comparing shaping triples using the orders in Definition 6. Both forms of monotonicity are parametric w.r.t.  $*$ , and strict monotonicity is also parametric w.r.t.  $\ll$ .

**Principle 2.** A strength function  $\sigma$  is:

- *monotonic* iff for any  $\alpha, \beta \in \mathcal{X}$ , if  $ST(\alpha) \simeq_* ST(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  and if  $ST(\alpha) \leq_* ST(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ ;
- *strictly monotonic* iff  $\sigma$  is monotonic and for any  $\alpha, \beta \in \mathcal{X}$ , if  $ST(\alpha) <_* ST(\beta)$  then  $\sigma(\alpha) \ll \sigma(\beta)$ .

Note that if  $\ll = \leq$ , then strict monotonicity coincides with monotonicity. For illustration, consider the QBAF in Fig. 1 with  $\tau(a) = \tau(b)$  and  $\sigma(d) = \perp$ . For a strictly monotonic  $\sigma$ :

- if  $* = \sigma \perp$ , by Definitions 3 and 6,  $ST(a) <_* ST(b)$ . Then, by Principle 2,  $\sigma(a) \ll \sigma(b)$ . For  $\ll = <$ ,  $\sigma(a) \ll \sigma(b)$  amounts to  $\sigma(a) < \sigma(b)$ , while for  $\ll = <_{\perp}$ , it amounts to  $\sigma(a) < \sigma(b)$  unless  $\sigma(a) = \sigma(b) = \perp$ ;
- if  $* = \sigma \perp$ , by Definitions 3 and 6,  $ST(a) \simeq_* ST(b)$  and then, by Principle 2,  $\sigma(a) = \sigma(b)$ .

Note that for  $\ll = <_{\perp}$  ( $\ll = <_{\top}$ ), if a shaping triple already gives rise to a  $\perp$  ( $\top$ , resp.) strength, a less (more, resp.) boosting triple cannot produce a strictly lesser (greater, resp.) strength. For  $\ll = <$ , strict monotonicity and the attainment of extreme values are somehow incompatible.

We foresee different applications requiring strict monotonicity, monotonicity or neither of these principles. Strict monotonicity seems to suit applications in which monotonic behaviour is always required. An example of this may be weighting the comments in social media (see [10]), in which every attacking or supporting comment, no matter how weak, may have an effect on an argument's score to maintain users' interests. For example, consider the QBAF shown in Fig. 3 in which  $\mathcal{X} = \{a, b, c, d\}$  with  $\mathcal{R}^-(a) = \{b, c\}$  and  $\mathcal{R}^+(a) = \{d\}$ . Strict monotonicity here may correspond to the addition of  $c$  decreasing  $\sigma(a)$  regardless of  $\sigma(a)$ ,  $\sigma(b)$  and  $\sigma(d)$  beforehand. However, this may not be desirable behaviour in settings where the aggregated strengths of an argument's attackers or supporters may require saturation. An example of this may be a semantics deployed in an engineering discussion on a component's design (see [20]) where an attacker of an argument proposing a particular design may be that said design would cause the component to fail. Here, this attacker may take on a special "defeater" status, e.g.  $\sigma(b) = \top$ , which maximises the aggregated attacking component of  $a$ , irrespective of its other attackers, and therefore the addition of  $c$  has no effect on  $\sigma(a)$ . Finally, some applications require that monotonicity is not fulfilled. These may include settings, such as political debates, in which self-attacks are more powerful than other attacks, regardless of the attackers' strengths (see [5]). For example, if the attacks from  $b$  and  $c$ , where  $\sigma(b) > \perp$  or  $\sigma(c) > \perp$ , were replaced by a self-attack from  $a$ , then  $\mathcal{R}^-(a)$  is reduced based on any choice of  $*$  but it may also be desirable that  $\sigma(a)$  is also reduced, due to the contradiction the self-attack represents.

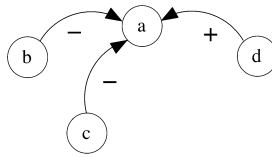


Fig. 3. Example QBAF for monotonicity.

Table 4

GPs implied by the principles.

|                     | GP1 | GP2 | GP3 | GP4 | GP5 | GP6 | GP7 | GP8 | GP9 | GP10 | GP11 |
|---------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| Balance             | ✓   | ✓   | ✓   |     |     |     |     |     |     |      |      |
| Strict Balance      | ✓   | ✓   | ✓   | ✓   | ✓   |     |     |     |     |      |      |
| Monotonicity        |     |     |     |     |     | ✓   |     |     |     |      |      |
| Strict Monotonicity |     |     |     |     |     | ✓   | ✓   | ✓   | ✓   | ✓    | ✓    |

**Proposition 19.** If  $\sigma$  is monotonic for some  $*$ ,  $\ll$  then it satisfies GP6 for the same  $*$ ,  $\ll$ ; if  $\sigma$  is strictly monotonic for some  $*$ ,  $\ll$  then it satisfies GP6 to GP11 for the same  $*$ ,  $\ll$ .

Note that the reverse of Proposition 19 does not hold. For example, if  $a, b$  are arguments in a QBAF such that  $\mathcal{R}^-(b) <_* \mathcal{R}^-(a)$ ,  $\mathcal{R}^+(b) >_* \mathcal{R}^+(a)$  and  $\tau(b) > \tau(a)$  but  $\sigma(a) > \sigma(b)$  then GP6–GP11 may hold while  $\sigma$  is not (strictly) monotonic.

Note also that other properties have been considered in the literature which are directly implied by (strict) monotonicity, as shown by the following propositions.

**Proposition 20.** Monotonicity implies the following:

*Bi-variate Monotony* [15] with  $*$  =  $\sigma \perp$  – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta) > \perp$ ,  $\mathcal{R}^-(\alpha) \subseteq \mathcal{R}^-(\beta)$  and  $\mathcal{R}^+(\beta) \subseteq \mathcal{R}^+(\alpha)$ , then  $\sigma(\alpha) \geq \sigma(\beta)$ .

*Bi-variate Reinforcement* [15] with  $*$  =  $\sigma \perp$  – For any  $\alpha, \beta \in \mathcal{X}$ ,  $\forall S, T \subseteq \mathcal{X}$  and  $\forall \gamma, \delta, \epsilon, \zeta \in \mathcal{X} \setminus (S \cup T)$ , if  $\tau(\alpha) = \tau(\beta) > \perp$ ,  $\mathcal{R}^-(\alpha) = S \cup \{\gamma\}$ ,  $\mathcal{R}^-(\beta) = S \cup \{\delta\}$ ,  $\mathcal{R}^+(\alpha) = T \cup \{\epsilon\}$ ,  $\mathcal{R}^+(\beta) = T \cup \{\zeta\}$ ,  $\sigma(\gamma) \leq \sigma(\delta)$  and  $\sigma(\epsilon) \geq \sigma(\zeta)$ , then  $\sigma(\alpha) \geq \sigma(\beta)$ .

**Proposition 21.** Strict monotonicity implies the following:

*Strict Bi-variate Monotony* [15] with  $*$  =  $\sigma \perp$  and  $\ll = <_{\perp, \top}$  – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta) > \perp$ ,  $\mathcal{R}^-(\alpha) \subseteq \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\beta) \subseteq \mathcal{R}^+(\alpha)$ ,  $(\sigma(\alpha) > 0$  or  $\sigma(\beta) < 1)$  and  $(\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$  or  $\mathcal{R}_{\sigma \perp}^+(\beta) \subsetneq \mathcal{R}_{\sigma \perp}^+(\alpha)$ ), then  $\sigma(\alpha) > \sigma(\beta)$ .

*Strict Bi-variate Reinforcement* [15]  $*$  =  $\sigma \perp$  and  $\ll = <_{\perp, \top}$  – For any  $\alpha, \beta \in \mathcal{X}$ ,  $\forall S, T \subseteq \mathcal{X}$  and  $\forall \gamma, \delta, \epsilon, \zeta \in \mathcal{X} \setminus (S \cup T)$ , if  $\tau(\alpha) = \tau(\beta) > \perp$ ,  $\mathcal{R}^-(\alpha) = S \cup \{\gamma\}$ ,  $\mathcal{R}^-(\beta) = S \cup \{\delta\}$ ,  $\mathcal{R}^+(\alpha) = T \cup \{\epsilon\}$ ,  $\mathcal{R}^+(\beta) = T \cup \{\zeta\}$ ,  $\sigma(\gamma) \leq \sigma(\delta)$ ,  $\sigma(\epsilon) \geq \sigma(\zeta)$ ,  $(\sigma(\alpha) > 0$  or  $\sigma(\beta) < 1)$  and  $(\sigma(\gamma) < \sigma(\delta)$  or  $\sigma(\epsilon) > \sigma(\zeta))$ , then  $\sigma(\alpha) > \sigma(\beta)$ .

Table 4 summarises the main results in this section.

## 5. Principles versus existing approaches

In this section we show that suitable instances (i.e. with assigned type of QBAF,  $\mathbb{I}$ ,  $*$  and  $\ll$ , and possibly fixed base score) of the principles introduced in Section 4 are satisfied by several gradual evaluation methods considered in the literature for various kinds of argumentation framework, as well as by some non-gradual semantics for AFs. Thus, by virtue of Propositions 17, 19, we synthetically prove that each approach satisfies the implied GPs and any literature and other property they are equivalent to or that the GPs imply. This shows that the level of generalisation afforded by the principles is well-chosen, by showing that the principles, for specific choices of parameters, are satisfied by existing concrete gradual and non-gradual semantics for various instances of QBAFs.

The results obtained in this section are summarised in Table 5. Here, we indicate with a ✓ the instances of our principles that hold and with a · the instances that do not hold (as illustrated by suitable counter-examples later). If the choices are indicated with a –, that choice is not applicable to the semantics, e.g. if a semantics does not hold balance or strict monotonicity then the choice for  $\ll$  is not applicable. The column labelled  $\tau$  gives the fixed base score for the QBAFs where this is present.

Note that if a principle is satisfied for  $\ll = <$  then it is also satisfied for every possible  $\ll$  since  $<$  is the strongest possible instance of  $\ll$ .

This table also highlights alternative semantics which may be interesting but have not yet been proposed in the literature. For example, a new form of Social Models semantics which suited the choice  $*$  =  $\sigma \perp$  would exhibit different behaviour to the original semantics and therefore may satisfy fewer properties and be suitable for different applications.

**Table 5**

Semantics in the literature assessed against our principles: Balance (B), Strict Balance (SB), Monotonicity (M) and Strict Monotonicity (SM).

| Semantics                 | Choices |                |        |                |                   | Principles |    |   |    |
|---------------------------|---------|----------------|--------|----------------|-------------------|------------|----|---|----|
|                           | QBAF    | $\mathbb{I}$   | $\tau$ | $*$            | $\ll$             | B          | SB | M | SM |
| h-Categorizer             | aQBAF   | (0, 1]         | 1      | $\sigma \perp$ | $<$               | ✓          | ✓  | ✓ | ✓  |
| Inverse Equational System | aQBAF   | [0, 1]         | 1      | $\sigma \perp$ | $<_{\perp}$       | ✓          | ✓  | ✓ | ✓  |
| Suspect Equational System | aQBAF   | [0, 1]         | 1      | $\sigma \perp$ | $\leq$            | ✓          | ✓  | . | .  |
| Max Equational System     | aQBAF   | [0, 1]         | 1      | $\sigma \perp$ | $<$               | ✓          | ✓  | ✓ | .  |
| GTAA                      | aQBAF   | [0, 1]         | 1      | $\sigma \perp$ | $<$               | ✓          | ✓  | . | .  |
| GTBA                      | QBAF    | [0, 1]         | 1      | –              | –                 | .          | .  | . | .  |
| LocMax                    | QBAF    | [−1, 1]        | 0      | $\sigma \perp$ | $<_0$             | ✓          | .  | ✓ | .  |
| LocSum                    | QBAF    | [−1, 1]        | 0      | $\sigma \perp$ | $<$               | ✓          | .  | ✓ | ✓  |
| Social Models             | aQBAF   | ordered set    | –      | $\sigma \perp$ | $<$               | ✓          | ✓  | ✓ | ✓  |
| Weighted Max-based        | aQBAF   | [0, 1]         | –      | $\sigma \perp$ | $<$               | ✓          | ✓  | ✓ | .  |
| Weighted Card-based       | aQBAF   | [0, 1]         | –      | $\sigma \perp$ | $<$               | ✓          | ✓  | ✓ | ✓  |
| Weighted h-Categorizer    | aQBAF   | [0, 1]         | –      | $\sigma \perp$ | $<$               | ✓          | ✓  | ✓ | ✓  |
| Top-Based Support         | sQBAF   | [0, 1]         | –      | $\sigma \perp$ | $<_{\top}$        | ✓          | ✓  | ✓ | .  |
| Reward-Based Support      | sQBAF   | [0, 1]         | –      | $\sigma \perp$ | $<_{\top}$        | ✓          | ✓  | ✓ | ✓  |
| Aggregation-Based Support | sQBAF   | [0, 1]         | –      | $\sigma \perp$ | $<_{\top}$        | ✓          | ✓  | ✓ | ✓  |
| QuAD                      | QBAF    | [0, 1]         | –      | $\sigma \perp$ | –                 | .          | .  | ✓ | .  |
| DF-QuAD                   | QBAF    | [0, 1]         | –      | $\sigma \perp$ | $\leq$            | ✓          | .  | ✓ | .  |
| Restricted Euler-based    | QBAF    | [0, 1]         | –      | $\sigma \perp$ | $<_{\perp, \top}$ | ✓          | .  | ✓ | ✓  |
| Complete Labellings       | aQBAF   | {in, out, und} | in     | $\sigma \perp$ | $<$               | ✓          | ✓  | ✓ | .  |

### 5.1. Besnard & Hunter categorizer

Besnard & Hunter [23] introduce the notion of *h-categoriser*, which basically is a function providing a quantitative evaluation of argument strength for tree-shaped abstract argumentation frameworks (aQBAFs with fixed base scores). In our notation, the h-categoriser is defined as follows, for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \frac{1}{1 + \sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta)} \quad (1)$$

Here, the fixed base score is 1 because for any  $\alpha \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$ , then  $\sigma(\alpha) = 1$ . It can be noted that  $0 < \sigma(\alpha) \leq 1$ , thus in this approach  $\mathbb{I} = (0, 1]$  with  $\top = 1$  and no bottom value  $\perp$  in  $\mathbb{I}$ . Consequently, in this approach there is no difference between  $\sigma \perp$  and  $\sigma \perp$ .

**Proposition 22.** *The h-categoriser is strictly balanced and strictly monotonic for any choice of  $*$  and  $\ll$  (i.e. for  $\ll = <$ ).*

To make the consequences of Proposition 22 explicit we note that this means that the h-categorizer satisfies the GPs implied by strict balance and strict monotonicity in virtue of Propositions 17 and 19 (i.e. GP1–GP11), and hence also all the properties implied (including the “hyphen” ones), according to Tables 2 and 3, by these GPs for aQBAFs with any choice of  $*$  and of  $\ll$ . As to GP1 this means that the h-categorizer satisfies Maximality (P1) and the hyphen in the same row for  $* = \sigma \perp$ . Similarly for other GPs (and their combinations in Table 3) it follows that the h-categorizer satisfies P5, P7, P9, P10, P15, P17, P18, P19, P20, P23, P25, P26, P27, P30, P32, P33 and the hyphens in the relevant rows.

### 5.2. Equational approaches

In [6] a generic numerical approach to argumentation semantics is proposed where each argument is assigned a value in the scale  $[0, 1]$  on the basis of a system of equations derived from the structure of the framework. Each equation in the system, called *basic equation* in the following, relates the value assigned to each argument with the values assigned to its attackers. In general the resulting system of equations may have many solutions, each representing a valid assignment of values to arguments.

We analyse in the following the three alternative forms of basic equation considered in [6]. Note that no notion of base score is considered in the equational approach and so for all three variants we assume that the fixed base score is 1 because for any  $\alpha \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$ , then  $\sigma(\alpha) = 1$ .

In the **Inverse Equational System** the basic equation for an argument  $\alpha$  is defined as follows:

$$\sigma(\alpha) = \prod_{\beta \in \mathcal{R}^-(\alpha)} (1 - \sigma(\beta)) \quad (2)$$

Note that any attacker  $\beta$  such that  $\sigma(\beta) = 0$  has no actual effect on  $\sigma(\alpha)$  hence the inverse equational system fits the option  $* = \sigma \perp$ .

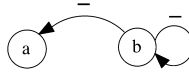


Fig. 4. Example (a)QBAF for Suspect Equational System and GTAA.

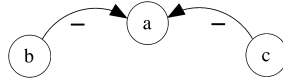


Fig. 5. Example aQBAF for Max Equational System and weighted max-based semantics.

**Proposition 23.** *The inverse equational system is strictly balanced and strictly monotonic with  $\ast = \sigma \perp$  and  $\ll = \leq$ .*

The **Suspect Equational System** is a variation of the inverse equational system where a special treatment is devoted to self-defeating arguments.

$$\sigma(\alpha) = \prod_{\beta \in \mathcal{R}^-(\alpha)} (1 - \sigma(\beta)) \quad \text{if } \alpha \notin \mathcal{R}^-(\alpha); \quad (3)$$

$$\sigma(\alpha) = \sigma(\alpha) \cdot \prod_{\beta \in \mathcal{R}^-(\alpha)} (1 - \sigma(\beta)) \quad \text{if } \alpha \in \mathcal{R}^-(\alpha). \quad (4)$$

This variation ensures that for a self-defeating  $\alpha$  argument it holds that  $\sigma(\alpha) = 0$ . As a consequence, one needs to consider  $\ast = \sigma \perp$  for (strict) balance since a self-defeating argument, namely a bottom strength attacker, reduces its own strength below its base score. In turn, we must also weaken the second parameter to  $\ll = \leq$  to account for bottom strength attackers not weakening an argument which is not self defeating.

**Proposition 24.** *The suspect equational system is strictly balanced with  $\ast = \sigma \perp$  and  $\ll = \leq$ .*

Compared with the inverse equational system the introduced variation breaks, as far as self-defeating arguments are concerned, monotonicity. Consider the aQBAF shown in Fig. 4 with fixed based score 1 in which  $\mathcal{X} = \{a, b\}$  with  $\mathcal{R}^-(a) = \{b\}$  and  $\mathcal{R}^-(b) = \{b\}$ . Here, monotonicity is shown not to hold for this semantics since  $ST(a) \simeq_\ast ST(b)$  but  $\sigma(a) = 1 \neq \sigma(b) = 0$ .

In the **Max Equational System** the basic equation for an argument  $\alpha$  is defined as follows:

$$\sigma(\alpha) = 1 - \max_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) \quad (5)$$

Here, any attacker  $\beta$  such that  $\sigma(\beta) = 0$  has no actual effect on  $\sigma(\alpha)$  hence the max equational system fits the option  $\ast = \sigma \perp$ .

**Proposition 25.** *The max equational system is strictly balanced and monotonic with  $\ast = \sigma \perp$  and with any  $\ll$ .*

It can be noted that strict monotonicity does not hold, since different sets of attackers having the same maximum value give rise to the same final score. For example, consider the aQBAF shown in Fig. 5 with fixed based score 1 in which  $\mathcal{X} = \{a, b, c\}$  with  $\mathcal{R}^-(a) = \{b, c\}$  and  $\sigma(b) = \sigma(c) = 1$ . Here, the max equational system gives  $\sigma(a) = 0$  both before and after  $c$  has been added to the framework.

### 5.3. Game-theoretic semantics

[5] and [22] give semantics for AFs and BFs using game-theoretic evaluations of arguments in two-person, zero-sum games. In both cases an argument's strength belongs to the interval  $\mathbb{I} = [0, 1]$  and corresponds to the value of a *game of argumentation strategy* when a proponent is forced to play that argument in its strategies (sets of arguments) against an opponent who may play any strategy.

We first consider the game-theoretic semantics in [5] for AFs, that we refer to as **GTAA**. We recall synthetically its definition in the following, the reader is referred to the original paper for more detailed explanations. For a given aQBAF  $\mathcal{Q}$  and  $\alpha \in \mathcal{X}$ , the sets of strategies for the proponent  $\mathcal{P}$  and opponent  $\mathcal{O}$  are  $S_{\mathcal{P}} = \{P \mid P \subseteq \mathcal{X}, \alpha \in P\}$  and  $S_{\mathcal{O}} = \{O \mid O \subseteq \mathcal{X}\}$ , respectively. The set of attacks from a set of arguments  $A \subseteq \mathcal{X}$  to a set of arguments  $B \subseteq \mathcal{X}$  is  $B \leftarrow^A = \{(a, b) \in A \times B \mid (a, b) \in \mathcal{R}^-\}$ . The *degree of acceptability function*  $\phi : S_{\mathcal{P}} \times S_{\mathcal{O}} \mapsto [0, 1]$  for a proponent strategy  $P \in S_{\mathcal{P}}$  relative to an opponent strategy  $O \in S_{\mathcal{O}}$  is defined as follows:

$$\phi(P, O) = \frac{1}{2} [1 + f(|O \leftarrow^P|) - f(|P \leftarrow^O|)] \quad (6)$$

where  $f(n) = \frac{n}{n+1}$ .

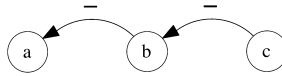


Fig. 6. Example (a)QBAF for GTAA.

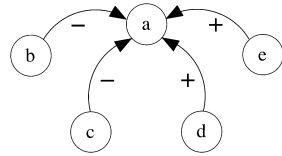


Fig. 7. Example QBAF for GTBA, QuAD, DF-QuAD, and restricted Euler-based semantics.

This becomes the *reward* for proponent  $\mathcal{P}$  whenever specified *borderline conditions* are not triggered. Formally, the reward for  $\mathcal{P}$  playing  $P$  against the opponent  $\mathcal{O}$  playing  $O$   $\rho: S_{\mathcal{P}} \times S_{\mathcal{O}} \mapsto [0, 1]$  is:

$$\rho(P, O) = \begin{cases} 0 & \text{iff } P \leftarrow P \neq \emptyset \\ 1 & \text{iff } P \leftarrow O = \emptyset \\ \phi(P, O) & \text{otherwise} \end{cases}$$

Let  $S_{\mathcal{P}} = \{P_1, P_2, \dots, P_m\}$  and  $S_{\mathcal{O}} = \{O_1, O_2, \dots, O_n\}$  with probability distributions  $p = \{p_1, p_2, \dots, p_m\}$  and  $o = \{o_1, o_2, \dots, o_n\}$ , respectively. For an argument  $\alpha \in \mathcal{X}$ , the strength  $\sigma(\alpha)$  is then the value of the game for the proponent playing  $\alpha$  using the minimax theorem, i.e.  $\sigma(\alpha) = \max_p \min_o E(\alpha, p, o)$ , where  $E(\alpha, p, o) = \sum_{j=1}^n \sum_{i=1}^m p_i o_j \rho(P_i, O_j)$  is the expected payoff.

Here, we see that attackers with  $\perp$  strength may have an effect on the attacked argument's strength, for example if they trigger one of the borderline conditions, and thus this approach fits the option  $\ast = \sigma \perp$ . We can also see that the fixed base score is 1 because for any  $\alpha \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$ , then  $\sigma(\alpha) = 1$ .

**Proposition 26.** *The GTAA semantics is strictly balanced with  $\ast = \sigma \perp$  and  $\ll = \leq$ .*

The GTAA semantics is not monotonic. For example, first consider the aQBAF shown in Fig. 4 with fixed base score 1 in which  $\mathcal{X} = \{a, b\}$  with  $\mathcal{R}^-(a) = \{b\}$  and  $\mathcal{R}^-(b) = \{b\}$ , where  $\sigma(a) = 0.25$  and  $\sigma(b) = 0$ . Then consider the aQBAF shown in Fig. 6 with fixed base score 1 in which  $\mathcal{X} = \{a, b, c\}$  with  $\mathcal{R}^-(a) = \{b\}$  and  $\mathcal{R}^-(b) = \{c\}$ , where  $\sigma(a) = 0.5$ ,  $\sigma(b) = 0.25$  and  $\sigma(c) = 1$ . From the former to the latter, the attackers of  $a$  increased in strength yet  $a$ 's strength has also increased.

We now consider the *asymmetrical* game-theoretic semantics for BFs in [22], which we refer to as **GTBA**. For a given QBAF  $\mathcal{Q}$  and  $\alpha \in \mathcal{X}$ , the sets of strategies for the proponent  $\mathcal{P}$  and opponent  $\mathcal{O}$  are defined similarly to the GTAA semantics. The set of supports from a set of arguments  $A \subseteq \mathcal{X}$  to another  $B \subseteq \mathcal{X}$  is  $B \Leftarrow A = \{(a, b) \in A \times B \mid (a, b) \in \mathcal{R}^+\}$ . The degree of acceptability function  $\phi: S_{\mathcal{P}} \times S_{\mathcal{O}} \mapsto [0, 1]$  for a proponent strategy  $P$  relative to an opponent strategy  $O$ , where  $f(n) = \frac{n}{n+1}$ , is:

$$\phi(P, O) = \frac{1}{4} [2 + f(|O \Leftarrow P|) - f(|P \Leftarrow O|) + f(|P \Leftarrow P \cup O|) - f(|O \Leftarrow P \cup O|)] \quad (7)$$

The strength is then calculated using the reward function in an identical manner to the GTAA semantics.

Once again, attackers with  $\perp$  strength may have an effect on the attacked argument's strength, as may supporters with  $\perp$  strength on a supported argument's strength, for example if they trigger one of the borderline conditions, and thus this approach also fits the option  $\ast = \sigma \perp$ . We can also see that the fixed base score is 1 because for any  $\alpha \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^+(\alpha) = \emptyset$ , then  $\sigma(\alpha) = 1$ .

The GTBA semantics is not balanced. For example, consider the QBAF shown in Fig. 7 with fixed base score 1 in which  $\mathcal{X} = \{a, b, c, d, e\}$  with  $\mathcal{R}^-(a) = \{b, c\}$  and  $\mathcal{R}^+(a) = \{d, e\}$ . Here,  $\sigma(a) = 0.375$  even though  $a$ 's attackers and supporters are balanced, since attacks are of higher importance than supports in this semantics due to its adversarial nature.

The GTBA semantics is not monotonic. For example, first consider the QBAF shown in Fig. 4 with fixed base score 1 in which  $\mathcal{X} = \{a, b\}$  with  $\mathcal{R}^-(a) = \{b\}$  and  $\mathcal{R}^-(b) = \{b\}$ , where  $\sigma(a) = 0.375$  and  $\sigma(b) = 0$ . Then consider the QBAF shown in Fig. 6 with fixed base score 1 in which  $\mathcal{X} = \{a, b, c\}$  with  $\mathcal{R}^-(a) = \{b\}$  and  $\mathcal{R}^-(b) = \{c\}$ , where  $\sigma(a) = 0.5$ ,  $\sigma(b) = 0.375$  and  $\sigma(c) = 1$ . From the former to the latter, the attackers of  $a$  increased in strength yet  $a$ 's strength has also increased, while its supporters were empty throughout.

In most cases both game-theoretical semantics behave *monotonically*; it is exceptions like the one shown in the counter-examples that cause monotonicity to not be satisfied. This arises from the fact that on the opponent side there is no counterpart of the “conflict-free” borderline conditions, to which the proponent is required to adhere, in either semantics.

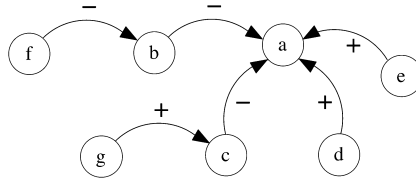


Fig. 8. Example QBAF for LocMax.

Choosing different borderline conditions and acceptability functions, for example following the spirit of those which are labelled as *symmetrical* in [22],<sup>12</sup> would possibly allow the satisfaction of the principles.

#### 5.4. Local valuation in bipolar frameworks

Amgoud et al. [2] introduce a generic local gradual evaluation scheme for arguments in bipolar argumentation frameworks (QBAFs with fixed base scores): we analyse here two instances thereof. In both cases arguments' strengths range over the interval  $\mathbb{I} = [-1, 1]$ . No explicit notion of base score is given in [2]: for our purposes here, the fixed base score is 0 because for any  $\alpha \in \mathcal{X}$ , if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^+(\alpha) = \emptyset$ , then  $\sigma(\alpha) = 0$ . The first instance, which we call **LocMax**, is defined as follows, for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \frac{\max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) - \max_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma)}{2} \quad (8)$$

Under the convention that when applied to the empty set the max operator returns  $\perp = -1$ , we get that supporters and attackers with  $\perp$  strength have no effect, and thus this approach fits the option  $\ast = \sigma \perp$ . To satisfy balance it also requires the choice  $\ll = <_0$ , where  $m <_0 n$  iff  $m < n$  or  $m = n = 0$ .

**Proposition 27.** *LocMax is balanced and monotonic with  $\ast = \sigma \perp$  and  $\ll = <_0$ .*

LocMax is not strictly balanced since, for example,  $\sigma(\alpha) < \tau(\alpha)$  can occur when the maximum strength of the attackers is greater than that of the supporters but the attackers and supporters are incomparable for any choice of  $\ast$ . For example, consider the QBAF shown in Fig. 8 with fixed base score 0 in which  $\mathcal{X} = \{a, b, c, d, e, f, g\}$  with  $\mathcal{R}^-(a) = \{b, c\}$ ,  $\mathcal{R}^+(a) = \{d, e\}$ ,  $\mathcal{R}^-(b) = \{f\}$  and  $\mathcal{R}^+(c) = \{g\}$ , where  $\sigma(b) = -0.5$ ,  $\sigma(c) = 0.5$ ,  $\sigma(d) = 0$ ,  $\sigma(e) = 0$ ,  $\sigma(f) = 0$  and  $\sigma(g) = 0$ . Here, LocMax gives  $\sigma(a) = -0.25$  but  $\mathcal{R}^-(a)$  and  $\mathcal{R}^+(a)$  are incomparable for any choice of  $\ast$ . The same framework can be used to demonstrate a counter-example for strict monotonicity since LocMax gives  $\sigma(a) = -0.25$  both before and after  $e$  is added to the framework.

The second instance, called **LocSum**, is given, for  $\alpha \in \mathcal{X}$ , by:

$$\sigma(\alpha) = \frac{1}{1 + h(\mathcal{R}^-(\alpha))} - \frac{1}{1 + h(\mathcal{R}^+(\alpha))} \quad (9)$$

where, for  $S \subseteq \mathcal{X}$ ,  $h(S) \triangleq \sum_{\beta \in S} \frac{\sigma(\beta) + 1}{2}$ . Under the convention that when applied to the empty set the sum returns 0, we get that supporters and attackers with a strength of  $-1 = \perp$  have no effect, and thus this approach fits the option  $\ast = \sigma \perp$ .

**Proposition 28.** *LocSum is balanced and strictly monotonic with  $\ast = \sigma \perp$  and for any choice of  $\ll$  (i.e. with  $\ll = <$ ).*

LocSum is not strictly balanced since, for example,  $\sigma(a) < 0$  can occur when  $h(\mathcal{R}^-(a)) > h(\mathcal{R}^+(a))$  but the attackers and supporters are incomparable for any choice of  $\ast$ . For example, consider the QBAF shown in Fig. 9 with fixed base score 0 in which  $\mathcal{X} = \{a, b, c, d, e, f, g, h\}$  with  $\mathcal{R}^-(a) = \{b, c\}$ ,  $\mathcal{R}^+(a) = \{d, e\}$ ,  $\mathcal{R}^-(b) = \{f\}$ ,  $\mathcal{R}^+(c) = \{g\}$  and  $\mathcal{R}^-(f) = \{h\}$ , where  $\sigma(b) = -0.25$ ,  $\sigma(c) = 0.333$ ,  $\sigma(d) = 0$ ,  $\sigma(e) = 0$ ,  $\sigma(f) = -0.333$ ,  $\sigma(g) = 0$  and  $\sigma(h) = 0$ . Here, LocMax gives  $\sigma(a) = -0.01$  but  $\mathcal{R}^-(a)$  and  $\mathcal{R}^+(a)$  are incomparable for any choice of  $\ast$ .

#### 5.5. Leite & Martins social model

[10] introduce a generic gradual evaluation method for social abstract argumentation frameworks (aQBAFs). The method is defined in terms of *well-behaved social abstract argumentation semantic framework*, which is a 5-tuple  $S = (L, \tau, \wedge, \gamma, \neg)$  where:  $L$  is a totally ordered set including  $\top$  and  $\perp$ ;  $\tau$  is a vote aggregation function which assigns a base score in  $L$  to each argument  $\alpha$  based on the positive and negative votes received by  $\alpha$  in the considered social context; the operator

<sup>12</sup> These semantics are outside the scope of this work since they give a strength of an argument relative to another, rather than per se, and so were not assessed here.



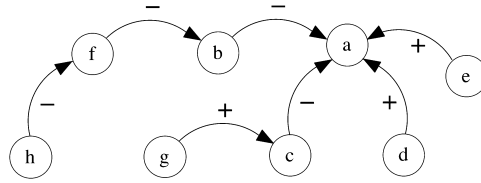


Fig. 9. Example QBAF for LocSum.

$\wedge : L \times L \rightarrow L$  is continuous, commutative, associative, monotonic w.r.t. both arguments, and  $\top$  is its identity element; the operator  $\vee : L \times L \rightarrow L$  is continuous, commutative, associative, monotonic w.r.t. both arguments, and  $\perp$  is its identity element; the operator  $\neg : L \rightarrow L$  is antimonotonic, continuous, with  $\neg\top = \perp$ ,  $\neg\perp = \top$ , and  $\neg\neg x = x$ . Then  $\sigma : \mathcal{X} \rightarrow L$  is a  $S$ -model iff for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \tau(\alpha) \wedge \neg \vee \{\sigma(\beta) \mid \beta \in \mathcal{R}^-(\alpha)\} \quad (10)$$

It can be noted that  $\perp$  is the identity element of  $\vee$  and thus this approach fits the option  $* = \sigma \perp$ . In [10] there is no remark about whether the monotonicity properties of  $\wedge$ ,  $\vee$ , and  $\neg$  are to be interpreted as strict. By interpreting them as strict then a well-behaved  $S$ -model is balanced.<sup>13</sup> Monotonicity is part of the definition of  $S$ -model, and is strict if the monotonicity properties of  $\wedge$ ,  $\vee$ , and  $\neg$  are strict. Formally:

**Proposition 29.** *A well behaved  $S$ -model is monotonic with  $* = \sigma \perp$ . If the monotonicity properties of  $\wedge$ ,  $\vee$ , and  $\neg$  are strict, then a well behaved  $S$ -model is strictly balanced and strictly monotonic with  $* = \sigma \perp$  and for any choice of  $\ll$  (i.e. for  $\ll = <$ ).*

### 5.6. Semantics for weighted argumentation

[14] gives three semantics for Weighted Argumentation Frameworks (aQBAFs), analysed below.

The **weighted max-based semantics** is such that (Theorems 4 and 5 of [14]) for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \frac{\tau(\alpha)}{1 + \max_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta)}. \quad (11)$$

Note that attackers with 0 strength have no actual effect in Eq. (11), and thus this approach fits the option  $* = \sigma \perp$ .

**Proposition 30.** *The weighted max-based semantics is strictly balanced and monotonic with  $* = \sigma \perp$  and for any choice of  $\ll$  (i.e. with  $\ll = <$ ).*

The weighted max-based semantics is not strictly monotonic as clearly there can be cases where  $ST(\alpha) <_{\sigma \perp} ST(\beta)$  but  $\tau(\alpha) = \tau(\beta)$  and  $\max_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma) = \max_{\delta \in \mathcal{R}^-(\beta)} \sigma(\delta)$ . For example, consider the aQBAF shown in Fig. 5 in which  $\mathcal{X} = \{a, b, c\}$  with  $\mathcal{R}^-(a) = \{b, c\}$  and  $\tau(a) = \tau(b) = \tau(c) = 1$ . Here, the weighted max-based semantics gives  $\sigma(a) = 0.5$  both before and after  $c$  has been added to the framework.

The **weighted card-based semantics** is such that (Theorems 6 and 7 of [14]) for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \frac{\tau(\alpha)}{1 + |\mathcal{R}^-(\alpha)| + \frac{\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta)}{|\mathcal{R}^-(\alpha)|}}, \quad (12)$$

where  $\frac{\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta)}{|\mathcal{R}^-(\alpha)|} = 0$  if  $|\mathcal{R}^-(\alpha)| = 0$ .

Note that attackers with 0 strength play a role in (12) (e.g. through the term  $|\mathcal{R}^-(\alpha)|$ ), and thus for this approach  $* = \sigma \perp$ .

**Proposition 31.** *The weighted card-based semantics is strictly balanced and strictly monotonic for  $* = \sigma \perp$  and any  $\ll$ .*

The **weighted h-categorizer semantics** generalises the h-categorizer given in Section 5.1 so that, for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \frac{\tau(\alpha)}{1 + \sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta)}. \quad (13)$$

<sup>13</sup> This interpretation is natural as non-strict monotonicity might lead to flat evaluation methods annihilating all differences. The instance in [10] satisfies strict monotonicity.

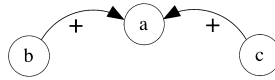


Fig. 10. Example sQBAF for top-based semantics.

Note that attackers with 0 strength have no actual effect in Eq. (13), and thus this approach fits the option  $\ast = \sigma \perp$ .

**Proposition 32.** *The weighted  $h$ -categoriser is strictly balanced and strictly monotonic with  $\ast = \sigma \perp$  and for any  $\ll$ .*

### 5.7. Amgoud & Ben-Naim support semantics

[3] introduce three gradual semantics for support argumentation frameworks (sQBAFs).

According to the **top-based semantics** the strength of arguments ranges in  $\mathbb{I} = [0, 1]$  and is related to their base score and to the strength of their strongest supporter (see Theorem 3 of [3]) according to the following equation, for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \tau(\alpha) + (1 - \tau(\alpha)) \max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) \quad (14)$$

where  $\max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) = 0$  if  $\mathcal{R}^+(\alpha) = \emptyset$ . Accordingly, supporters with  $\perp$  strength have no effect, which means that this approach fits the option  $\ast = \sigma \perp$ .

**Proposition 33.** *The top-based semantics is strictly balanced and monotonic with  $\ast = \sigma \perp$  and  $\ll = \leq_{\top}$ .*

The top-based semantics is not strictly monotonic since clearly there can be cases where  $\mathcal{ST}(a) <_{\sigma \perp} \mathcal{ST}(b)$  but  $\tau(a) = \tau(b)$  and  $\max_{c \in \mathcal{R}^+(a)} \sigma(c) = \max_{d \in \mathcal{R}^+(b)} \sigma(d)$ . For example, consider the sQBAF shown in Fig. 10 in which  $\mathcal{X} = \{a, b, c\}$  with  $\mathcal{R}^+(a) = \{b, c\}$  and  $\tau(a) = \tau(b) = \tau(c) = 0.5$ . Here, top-based semantics gives  $\sigma(a) = 0.75$  both before and after  $c$  has been added to the framework.

The **reward-based semantics** is based on the notion of *founded* argument:  $\alpha \in \mathcal{X}$  is founded iff there is a finite sequence  $\alpha_0 \dots \alpha_n$ , for  $n \geq 0$ , such that  $\tau(\alpha_0) > 0$ ,  $\alpha_n = \alpha$ , and for every  $i = 0, \dots, n-1$ :  $\alpha_i \in \mathcal{R}^+(\alpha_{i+1})$ . Then, the strength of arguments ranges in  $\mathbb{I} = [0, 1]$  and is related to their base score and to the strength of their supporters (see Theorem 6 of [3]) according to the following equation, for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \tau(\alpha) + (1 - \tau(\alpha)) \left( \sum_{j=1}^{n(\alpha)-1} \frac{1}{2^j} + \frac{m(\alpha)}{2^{n(\alpha)}} \right) \quad (15)$$

where  $n(\alpha) = |\mathcal{R}^+(\alpha)|$  and  $m(\alpha) = \frac{\sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta)}{n(\alpha)}$  and, by convention,  $\sum_{j=1}^{n(\alpha)-1} \frac{1}{2^j} + \frac{m(\alpha)}{2^{n(\alpha)}} = 0$  if  $\mathcal{R}^+(\alpha) = \emptyset$ .

Proposition 4 of [3] proves that if an argument  $\alpha$  is founded then  $\sigma(\alpha) > 0$ . It is also immediate to see that if  $\sigma(\alpha) > 0$  then  $\alpha$  is founded (from  $\sigma(\alpha) > 0$  and Equation (15) we get that either  $\tau(\alpha) > 0$  or  $\mathcal{R}^+(\alpha) \neq \emptyset$ ). It follows that supporters with zero strength have no effect, and thus this approach fits the option  $\ast = \sigma \perp$ .

**Proposition 34.** *The reward-based semantics is strictly balanced and strictly monotonic with  $\ast = \sigma \perp$  and  $\ll = \leq_{\top}$ .*

Finally, according to the **aggregation-based semantics**, the strength of arguments ranges in  $\mathbb{I} = [0, 1]$  and is related to their base score and to the strength of their supporters (see Theorem 9 of [3]) according to the following equation, for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = \tau(\alpha) + (1 - \tau(\alpha)) \frac{\sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta)}{1 + \sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta)} \quad (16)$$

It follows that supporters with zero strength have no effect (they do not affect the sum  $\sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta)$ ), and thus this approach fits the option  $\ast = \sigma \perp$ .

**Proposition 35.** *The aggregation-based semantics is strictly balanced and strictly monotonic with  $\ast = \sigma \perp$  and  $\ll = \leq_{\top}$ .*

### 5.8. Semantics for QuAD frameworks

[9] and [20] give semantics for Quantitative Argumentation Debate (QuAD) Frameworks (QBAFs), analysed below. In both cases arguments' strengths are over the interval  $\mathbb{I} = [0, 1]$ . The DF-QuAD algorithm is a "discontinuity-free" development of the QuAD algorithm.

The **QuAD** algorithm [9] is defined as follows, for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = g(\tau(\alpha), \mathcal{F}_a(\tau(\alpha), \sigma(\mathcal{R}^-(\alpha))), \mathcal{F}_s(\tau(\alpha), \sigma(\mathcal{R}^+(\alpha)))) \quad (17)$$

where if  $(\alpha_1, \dots, \alpha_n)$  is an arbitrary permutation of the  $(n \geq 0)$  attackers in  $\mathcal{R}^-(\alpha)$ ,  $\sigma(\mathcal{R}^-(\alpha)) = (\sigma(\alpha_1), \dots, \sigma(\alpha_n))$  (and similarly for supporters).

The operator  $g$  is defined as  $g : \mathbb{I} \times \mathbb{I} \cup \{\text{nil}\} \times \mathbb{I} \cup \{\text{nil}\} \rightarrow \mathbb{I}$ , where for  $v_0, v_a, v_s \in \mathbb{I}$ :

$$g(v_0, v_a, v_s) = v_a \quad \text{if } v_s = \text{nil} \text{ and } v_a \neq \text{nil} \quad (18)$$

$$g(v_0, v_a, v_s) = v_s \quad \text{if } v_a = \text{nil} \text{ and } v_s \neq \text{nil} \quad (19)$$

$$g(v_0, v_a, v_s) = v_0 \quad \text{if } v_a = v_s = \text{nil} \quad (20)$$

$$g(v_0, v_a, v_s) = \frac{v_a + v_s}{2} \quad \text{otherwise} \quad (21)$$

Letting  $\times$  stand for either  $a$  or  $s$ , the operator  $\mathcal{F}_\times$  is defined as  $\mathcal{F}_\times : \mathbb{I}^* \rightarrow \mathbb{I}$ , where for  $S = (v_1, \dots, v_m) \in \mathbb{I}^*$ ,  $(w_1, \dots, w_n)$  is an arbitrary permutation of the non-zero elements in  $S$ <sup>14</sup>:

$$\text{if } n = 0 : \quad \mathcal{F}_\times(v_0, S) = \text{nil} \quad (22)$$

$$\text{if } n = 1 : \quad \mathcal{F}_\times(v_0, S) = f_\times(v_0, w) \quad (23)$$

$$\text{if } n > 1 : \quad \mathcal{F}_\times(v_0, (w_1, \dots, w_n)) = f_\times(\mathcal{F}_\times(v_0, (w_1, \dots, w_{n-1})), w_n) \quad (24)$$

with the *base expressions*  $f_\times : \mathbb{I} \times \mathbb{I} \rightarrow \mathbb{I}$  defined, for  $v_0, v \in \mathbb{I}$ , as:

$$f_a(v_0, v) = v_0 - v_0 \cdot v = v_0 \cdot (1 - v) \quad (25)$$

$$f_s(v_0, v) = v_0 + (1 - v_0) \cdot v = v_0 + v - v_0 \cdot v \quad (26)$$

In both this approach and the one that follows we see that supporters and attackers with  $\perp$  strength have no effect, and thus they fit the option  $*$  =  $\sigma \perp$ .

**Proposition 36.** *The QuAD algorithm is monotonic with  $*$  =  $\sigma \perp$ .*

Note that the QuAD algorithm is not balanced. For example, consider the QBAF shown in Fig. 7 in which  $\mathcal{X} = \{a, b, c, d, e\}$  with  $\mathcal{R}^-(a) = \{b, c\}$  and  $\mathcal{R}^+(a) = \{d, e\}$ , where  $\tau(a) = \tau(b) = \tau(c) = \tau(d) = \tau(e) = 1$ . Here, the QuAD algorithm gives  $\sigma(a) = 0.5$  even though its attackers and supporters are balanced. The same framework demonstrates that strict monotonicity does not hold since  $\sigma(a) = 0.5$  both before and after  $c$  or  $e$  are added to the framework.

The **DF-QuAD** algorithm [20] is defined as follows, for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = c(\tau(\alpha), \mathcal{F}'(\sigma(\mathcal{R}^-(\alpha))), \mathcal{F}'(\sigma(\mathcal{R}^+(\alpha)))) \quad (27)$$

where if  $(\alpha_1, \dots, \alpha_n)$  is an arbitrary permutation of the  $(n \geq 0)$  attackers in  $\mathcal{R}^-(\alpha)$ ,  $\sigma(\mathcal{R}^-(\alpha)) = (\sigma(\alpha_1), \dots, \sigma(\alpha_n))$  (and similarly for supporters).

The *polarity function* is defined as  $c : \mathbb{I} \times \mathbb{I} \times \mathbb{I} \rightarrow \mathbb{I}$ , where for  $v_0, v_a, v_s \in \mathbb{I}$ :

$$c(v_0, v_a, v_s) = v_0 - v_0 \cdot |v_s - v_a| \quad \text{if } v_a \geq v_s \quad (28)$$

$$c(v_0, v_a, v_s) = v_0 + (1 - v_0) \cdot |v_s - v_a| \quad \text{if } v_a < v_s \quad (29)$$

The *strength aggregation function* is defined as  $\mathcal{F}' : \mathbb{I}^* \rightarrow \mathbb{I}$ , where for  $S = (v_1, \dots, v_n) \in \mathbb{I}^*$ :

$$\text{if } n = 0 : \mathcal{F}'(S) = 0 \quad (30)$$

$$\text{if } n = 1 : \mathcal{F}'(S) = v_1 \quad (31)$$

$$\text{if } n = 2 : \mathcal{F}'(S) = f'(v_1, v_2) \quad (32)$$

$$\text{if } n > 2 : \mathcal{F}'(S) = f'(\mathcal{F}'(v_1, \dots, v_{n-1}), v_n) \quad (33)$$

with the *base function*  $f' : \mathbb{I} \times \mathbb{I} \rightarrow \mathbb{I}$  defined, for  $v_1, v_2 \in \mathbb{I}$ , as:

$$f'(v_1, v_2) = v_1 + (1 - v_1) \cdot v_2 = v_1 + v_2 - v_1 \cdot v_2 \quad (34)$$

<sup>14</sup> This formulation is a modification of the original formulation of  $\mathcal{F}_\times$ , in which  $(w_1, \dots, w_n)$  was not used.

**Proposition 37.** *The DF-QuAD algorithm is balanced and monotonic with  $* = \sigma \perp$  and  $\ll = \leq$ .*

The DF-QuAD algorithm is not strictly balanced. For example, consider the QBAF shown in Fig. 7 in which  $\mathcal{X} = \{a, b, c, d, e\}$  with  $\mathcal{R}^-(a) = \{b, c\}$  and  $\mathcal{R}^+(a) = \{d, e\}$ , where  $\tau(a) = 0.5$ ,  $\tau(b) = 1$ ,  $\tau(c) = 0.2$ ,  $\tau(d) = 0.7$  and  $\tau(e) = 0.5$ . Here, the DF-QuAD algorithm gives  $\sigma(a) = 0.425 \neq \tau(a)$ ,  $\mathcal{R}^-(a)$  and  $\mathcal{R}^+(a)$  are incomparable for any choice of  $*$ . The same framework demonstrates that strict monotonicity does not hold since  $\sigma(a) = 0.425$  both before and after  $c$  has been added to the framework.

Both semantics for QuAD Frameworks do not satisfy the strict monotonicity property because the strength aggregation function saturates at the  $\top$  value.

### 5.9. Restricted Euler-based semantics

In [15] an Euler-based semantics for restricted (i.e. acyclic) BAG Frameworks (acyclic QBAFs) is introduced.

The **restricted Euler-based semantics** is such that for  $\alpha \in \mathcal{X}$ :

$$\sigma(\alpha) = 1 - \frac{1 - \tau(\alpha)^2}{1 + \tau(\alpha) \cdot e^E} \quad (35)$$

where:

$$E = \sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) - \sum_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma). \quad (36)$$

Here,  $0 \leq \sigma(\alpha) \leq 1$ , thus in this approach  $\mathbb{I} = [0, 1]$  with  $\top = 1$  and  $\perp = 0$ . It can be seen that attackers with 0 strength have no effect in Equations (35) and (36) and so this approach fits the option  $* = \sigma \perp$ . This semantics also exhibits behaviour such that for any  $\alpha \in \mathcal{X}$ , if  $\tau(\alpha) = 0$  then  $\sigma(\alpha) = 0$  or if  $\tau(\alpha) = 1$  then  $\sigma(\alpha) = 1$ . This therefore requires a new choice to be defined such that  $\ll = <_{\perp, \top}$ , where  $m <_{\perp, \top} n$  iff  $m < n$  or  $m = n = \top$  or  $m = n = \perp$ .

**Proposition 38.** *The restricted Euler-based semantics is balanced and strictly monotonic with  $* = \sigma \perp$  and  $\ll = <_{\perp, \top}$ .*

The restricted Euler-based semantics is not strictly balanced. For example, consider the QBAF shown in Fig. 7 in which  $\mathcal{X} = \{a, b, c, d, e\}$  with  $\mathcal{R}^-(a) = \{b, c\}$  and  $\mathcal{R}^+(a) = \{d, e\}$  and base scores  $\tau(a) = 0.5$ ,  $\tau(b) = 0.1$ ,  $\tau(c) = 0.9$ ,  $\tau(d) = 0.2$  and  $\tau(e) = 0.3$ . Here,  $\sigma(a) = 0.425 < \tau(a)$  even though it is not the case that  $\mathcal{R}^-(a) >_{\sigma \perp} \mathcal{R}^+(a)$ .

### 5.10. Non-gradual semantics for abstract argumentation

**Complete Semantics** [1] plays a central role in traditional Dung's semantics for AFs. It is based on the notion of the *complete extension*, which, briefly, is a conflict-free set of arguments which consists of all the arguments it defends. Many other semantics (in particular stable, preferred, grounded, semi-stable and ideal semantics) are *complete-based*, i.e. the extensions that they sanction are a subset of the set of complete extensions. Thus, results which are proven for complete semantics directly hold for the many other complete-based semantics. An extension can be equivalently expressed as a three-valued labelling of the arguments, using the set of labels  $\Lambda = \{\text{in}, \text{out}, \text{und}\}$ , where the label in for an argument corresponds to being a member of the extension, out to being attacked by the extension, und to not being a member nor being attacked. In terms of acceptance, these three values correspond to a simple qualitative scale  $\mathbb{I} = \{\text{in}, \text{und}, \text{out}\}$ , where  $\text{out} < \text{und} < \text{in}$ , so out is the bottom value  $\perp$  and in is the top value  $\top$ .

In this context, complete labellings can be formally characterised as follows. Given an AF  $\langle \mathcal{X}, \mathcal{R}^- \rangle$  a labelling  $L : \mathcal{X} \rightarrow \{\text{in}, \text{out}, \text{und}\}$  is a complete labelling for any  $\alpha \in \mathcal{X}$  [4]:

- $L(\alpha) = \text{in}$  iff  $\forall \beta \in \mathcal{R}^-(\alpha) : L(\beta) = \text{out}$ ;
- $L(\alpha) = \text{out}$  iff  $\exists \beta \in \mathcal{R}^-(\alpha) : L(\beta) = \text{in}$ ;
- $L(\alpha) = \text{und}$  iff  $\nexists \beta \in \mathcal{R}^-(\alpha) : L(\beta) = \text{in}$  and  $\exists \beta \in \mathcal{R}^-(\alpha) : L(\beta) = \text{und}$ .

Given that an unattacked argument is labelled in, it is natural to adopt the convention that  $\tau(\alpha) = \text{in}$  for every argument  $\alpha$ . Thus, an AF corresponds to an aQBAF with fixed base score in as indicated in Table 5. Further, it can be observed that arguments labelled out have no effective role in determining the labels of other arguments, hence complete labellings fit the option  $* = \sigma \perp$ . Complete labellings satisfy balance and monotonicity for aQBAFs with fixed base score in and  $\mathbb{I} = \{\text{in}, \text{und}, \text{out}\}$  as given above.

**Proposition 39.** *Every complete labelling is strictly balanced and monotonic with  $* = \sigma \perp$  and with any  $\ll$ .*

Note that the complete semantics does not satisfy strict monotonicity since there can be cases where  $ST(\alpha) <_* ST(\beta)$  while  $L(\alpha) = L(\beta)$ . In particular, the addition of attackers labelled und does not affect the status of an argument already

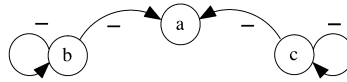


Fig. 11. Example aQBAF for complete semantics.

labelled und. For example, consider the aQBAF shown in Fig. 11 with fixed base score in, in which  $\mathcal{X} = \{a, b, c\}$  where  $\mathcal{R}^-(a) = \{b, c\}$ ,  $\mathcal{R}^-(b) = \{b\}$ ,  $\mathcal{R}^-(c) = \{c\}$ ,  $L(b) = \text{und}$  and  $L(c) = \text{und}$ . Here, the complete semantics gives  $L(a) = \text{und}$  both before and after  $c$  has been added to the framework.

**Corollary 1.** Any semantics for abstract argumentation that is complete-based is also strictly balanced and monotonic with  $* = \sigma \perp$  and with any  $\ll$ .

## 6. Discussion and conclusions

This paper provides a contribution towards an enhanced understanding of the landscape of properties for gradual argumentation in several respects.

First, while a variety of fine-grained properties have been previously introduced and discussed in the literature, thanks to the notion of parametric group properties we have shown that many more fine-grained properties can be identified.

Second, the existence of so many properties, together with the relative numerosity of group properties, has evidenced the need for some higher-level synthesis, which has been achieved with the general principles of balance and monotonicity. They have been shown to support a simplified compact analysis of gradual argumentation formalisms on the basis of the implication relations holding with respect to the group properties. The results in Section 5 illustrate the practical value of these principles, by showing that a variety of gradual methods, conceived in different contexts, satisfy suitable instances of both balance and monotonicity. Our principles' simplicity and power, confirmed by these results, is one of the main advancements with respect to the previous literature. As powerful as they are, we remark that the general principles are not meant to replace finer levels of analysis, which of course are required when (some of) the principles do not hold, while some of their implied properties might hold anyway.

Third, the parametric nature of the group properties and of the principles turns out to provide a further advantage as it allows one to directly reveal and compare some basic design choices underlying different formalisms. These concern in particular the role of bottom strength arguments and the possibility of saturation in the assigned strength (evidenced by the use of  $<_{\times}$  operators) and/or in the aggregated effects of attackers and supporters (occurring when monotonicity is not strict).

Making these choices explicit can be useful to identify variants of existing approaches, derivable by changing some parameter, and to support an investigation about which combinations of the parameters  $*$  and  $\ll$  are more appropriate with respect to the needs of different argumentative contexts (e.g. epistemic vs. practical reasoning, monological or dialogical argumentation). This appears to be a significant direction of future work.

At a higher level of abstraction, principles ease the classification of gradual argumentation approaches in broad families with different underlying notions of strength. The study of these families represents another important line of future investigation. As an example, we only mention the case of GTBA and QuAD formalisms, where the absence of the balance property indicates that the base score has quite a different role than in other approaches. For instance, in QuAD a top base score is insensitive to the presence of supporters, while it can be decreased by the presence of attackers.

While the GPs and principles proposed in this paper are fairly comprehensive, it is appropriate to remark that there are properties considered in the literature which they do not cover. In particular our approach assumes as a basis for strength assessment the strength values of the attackers and supporters, respectively, and cannot capture properties where other kinds of factors are considered. Some examples of these properties include *cardinality precedence* [3,14], which prioritises the cardinality of attackers/supporters rather than their strengths; *quality precedence* [3,14], which prioritises the maximum strength of attackers/supporters rather than their combined effect; and *compensation* [3,14] and *franklin* [15], both of which consider some kind of interaction between attackers and supporters themselves, a dense problem which warrants studies in itself.

Some other properties in the literature are implied by our principles while not being covered by our GPs. For example, as shown in Section 4, weakening and strengthening [15] are implied by balance, while (strict) bi-variate monotony and (strict) reinforcement [15] are implied by (strict, resp.) monotony. Our GPs have been conceived to correspond to relatively simple ideas, while GPs defined for the above properties would be less immediate and would closely resemble the general principles. The fact that these properties are relatively few in number justifies our choice of GPs.

In [24] the concepts of “subsumption” and “incompatibility” between ranking-based argumentation principles, i.e. literature properties, are introduced. Although this work shares our intentions of simplifying axiomatic analysis of gradual evaluation methods, it differs from ours in that we create group properties which subsume a number of literature properties, before showing that principles subsume these group properties. In other words we carry out a systematic analysis at different grain levels, while in [24] relationships between existing fine-grained properties are considered.

Some quantitative semantics were not included in the analysis of Section 5 since their behaviour exhibits significantly different characteristics to those we analysed. For example, the *compensation-based semantics* in [8] and similarly

the *variable-depth propagation* semantics in [25] satisfy neither balance nor monotonicity, due to the fact that the strength of an argument increases with the number of its attackers, provided that they are in turn attacked. Meanwhile the *graded semantics* of [26] uses the cardinality of attackers and defenders, rather than their strengths, in order to calculate an argument's strength. Studies on the relationships and shared ideas between these works and our principles is another interesting line of future work.

More generally, the study of the applicability of our principles to different gradual evaluation schemes, for example that proposed in [27], based on aspects and votes, or to different kinds of abstract frameworks, e.g. with weighted argument relationships as in [28], or considering additional relationships with respect to attack and support, may provide further insights into the principles' flexibility. In this respect, we remark that the notion of shaping triple lends itself to be extended to a notion of shaping tuple, including other elements while keeping the basic formulation of the principle of monotonicity. A generalization in the same direction, involving a change to the formal structure of the current definition while preserving the underlying intuition, would need to be applied to the principle of balance.

Finally, given the foundational nature of the approach, connections with other related works at large are worth examining.

In the argumentation context, this may concern for instance a comparison with the properties of traditional non-gradual semantics (e.g. *conflict-freeness* or *reinstatement*).

More generally, relationships with approaches to qualitative decision making can be considered. For instance the work presented in [29], concerning the qualitative assessments of pros and cons, contains many basic ideas (e.g. those for *weak unanimity* and *status quo consistency*) which may be relevant to the study of gradual properties addressed in this paper.

## Acknowledgements

The first author was partially supported by the University of Brescia under the WAT\_CHALLENGE project. The third author was partially funded by the EPSRC project EP/P029558/1 ROAD2H. The authors are indebted to anonymous reviewers for their helpful and insightful comments.

## Appendix A. Proofs

**Lemma 1.** For any  $A, B \subseteq \mathcal{X}$  such that  $A \geq_* B$ ,  $A >_* B$  iff  $|B^*| < |A^*|$  or for every injective mapping  $f$  from  $B^*$  to  $A^*$   $\exists \beta \in B^* : \sigma(f(\beta)) \not\leq \sigma(\beta)$ .

**Proof.** Given  $A \geq_* B$ , to prove the statement we have to show that  $B \not\geq_* A$  iff  $|B^*| < |A^*|$  or for every injective mapping  $f$  from  $B^*$  to  $A^*$   $\exists \beta \in B^* : \sigma(f(\beta)) \not\leq \sigma(\beta)$ . Now, by Definition 3, when  $B \not\geq_* A$  there is no injective mapping  $f$  from  $A^*$  to  $B^*$  such that  $\forall \alpha \in A^*, \sigma(f(\alpha)) \geq \sigma(\alpha)$ , which happens iff the set of injective mappings from  $A^*$  to  $B^*$  is empty (i.e.  $|B^*| < |A^*|$ ) or every member of this set does not satisfy the required condition (i.e. for every injective mapping  $f$  from  $B^*$  to  $A^*$   $\exists \beta \in B^* : \sigma(f(\beta)) \not\leq \sigma(\beta)$ ).  $\square$

## Proposition 16.

- GP6 and GP7 with  $* = \sigma \perp$  and with any  $\ll$  imply:  
P30. Monotony [12], [5, Prop. 6] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^-(\alpha) \subseteq \mathcal{R}^-(\beta)$  then  $\sigma(\alpha) \geq \sigma(\beta)$ .
- GP6 and GP8 with  $* = \sigma \perp$  and with any  $\ll$  imply:  
P31. Monotony [3] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^+(\alpha) \subseteq \mathcal{R}^+(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .
- GP6 and GP10 with  $* = \sigma \perp$  and with any  $\ll$  imply:  
P32. Theorem 7 in [8] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and there exists an injective mapping  $f$  from  $\mathcal{R}^-(\alpha)$  to  $\mathcal{R}^-(\beta)$  such that  $\forall \gamma \in \mathcal{R}^-(\alpha), \sigma(f(\gamma)) \geq \sigma(\gamma)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .  
P33. Counter-Transitivity [7,13] – For any  $\alpha, \beta \in \mathcal{X}$ , if  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^-(\alpha) \leq_{\sigma \perp} \mathcal{R}^-(\beta)$ , then  $\sigma(\alpha) \geq \sigma(\beta)$ .

**Proof.** We first prove for  $\ll = <$  and then the rest follows.

Let  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$  then, by GP6,  $\sigma(\alpha) = \sigma(\beta)$ . Let  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$  then, by GP7,  $\sigma(\alpha) > \sigma(\beta)$ . Therefore, in both cases  $\sigma(\alpha) \geq \sigma(\beta)$  and GP6 and GP7 imply Monotony [12].

Let  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  then, by GP6,  $\sigma(\alpha) = \sigma(\beta)$ . Let  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  then, by GP8,  $\sigma(\alpha) < \sigma(\beta)$ . Therefore, in both cases  $\sigma(\alpha) \leq \sigma(\beta)$  and GP6 and GP8 imply Monotony [3].

Let  $\tau(\alpha) = \tau(\beta)$  and there exists an injective mapping  $f$  from  $\mathcal{R}^-(\alpha)$  to  $\mathcal{R}^-(\beta)$  such that  $\forall \gamma \in \mathcal{R}^-(\alpha), \sigma(f(\gamma)) \geq \sigma(\gamma)$  and therefore, by Definition 3,  $\mathcal{R}^-(\alpha) \leq_{\sigma \perp} \mathcal{R}^-(\beta)$ . If  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \mathcal{R}^-(\beta)$  then, by GP6,  $\sigma(\alpha) = \sigma(\beta)$ . If  $\mathcal{R}^-(\beta) \geq_{\sigma \perp} \mathcal{R}^-(\alpha)$  and  $\mathcal{R}^-(\alpha) \not\leq_{\sigma \perp} \mathcal{R}^-(\beta)$ , i.e.  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \mathcal{R}^-(\beta)$ , then, by GP10,  $\sigma(\alpha) > \sigma(\beta)$ . Therefore, in both cases  $\sigma(\alpha) \geq \sigma(\beta)$  and GP6 and GP10 imply Theorem 7 in [8].

It follows that GP6 and GP10 imply Counter-Transitivity.  $\square$

**Proposition 17.** If  $\sigma$  is balanced for some  $*$ ,  $\ll$  then it satisfies GP1 to GP3 for the same  $*$ ,  $\ll$ ; if  $\sigma$  is strictly balanced for some  $*$ ,  $\ll$  then it satisfies GP1 to GP5 for the same  $*$ ,  $\ll$ .



**Proof.** Let  $\sigma$  be balanced.

If  $\mathcal{R}_*^-(\alpha) = \emptyset$  and  $\mathcal{R}_*^+(\alpha) = \emptyset$  then  $\mathcal{R}^-(\alpha) =_* \mathcal{R}^+(\alpha)$  and, by Principle 1.1,  $\sigma(\alpha) = \tau(\alpha)$ , therefore balance implies GP1.

If  $\mathcal{R}_*^-(\alpha) \neq \emptyset$  and  $\mathcal{R}_*^+(\alpha) = \emptyset$  then  $\mathcal{R}^-(\alpha) >_* \mathcal{R}^+(\alpha)$  and, by Principle 1.2,  $\sigma(\alpha) \ll \tau(\alpha)$ , therefore balance implies GP2.

If  $\mathcal{R}_*^-(\alpha) = \emptyset$  and  $\mathcal{R}_*^+(\alpha) \neq \emptyset$  then  $\mathcal{R}^-(\alpha) <_* \mathcal{R}^+(\alpha)$  and, by Principle 1.3,  $\sigma(\alpha) \gg \tau(\alpha)$ , therefore balance implies GP3.

Let  $\sigma$  be strictly balanced.

If  $\sigma(\alpha) < \tau(\alpha)$  then, by Principle 1.5,  $\mathcal{R}^-(\alpha) >_* \mathcal{R}^+(\alpha)$ , therefore  $\mathcal{R}^-(\alpha) \neq \emptyset$  and strict balance implies GP4.

If  $\sigma(\alpha) > \tau(\alpha)$  then, by Principle 1.6,  $\mathcal{R}^-(\alpha) <_* \mathcal{R}^+(\alpha)$ , therefore  $\mathcal{R}^+(\alpha) \neq \emptyset$  and strict balance implies GP5.  $\square$

**Proposition 19.** If  $\sigma$  is monotonic for some  $*$ ,  $\ll$  then it satisfies GP6 for the same  $*$ ,  $\ll$ ; if  $\sigma$  is strictly monotonic for some  $*$ ,  $\ll$  then it satisfies GP6 to GP11 for the same  $*$ ,  $\ll$ .

**Proof.** Let  $\sigma$  be monotonic.

If  $\mathcal{R}^-(\alpha) =_* \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) =_* \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then, by Definition 6,  $ST(\alpha) \simeq_* ST(\beta)$ . Therefore, by Principle 2,  $\sigma(\alpha) = \sigma(\beta)$  and (strict) monotonicity implies GP6.

Let  $\sigma$  be strictly monotonic.

If  $\mathcal{R}_*^-(\alpha) \supseteq \mathcal{R}_*^-(\beta)$ ,  $\mathcal{R}_*^+(\alpha) = \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\mathcal{R}^-(\alpha) >_* \mathcal{R}^-(\beta)$  and, by Definition 6,  $ST(\alpha) <_* ST(\beta)$ . Therefore, by Principle 2,  $\sigma(\alpha) \ll \sigma(\beta)$  and strict monotonicity implies GP7.

If  $\mathcal{R}_*^-(\alpha) = \mathcal{R}_*^-(\beta)$ ,  $\mathcal{R}_*^+(\alpha) \supseteq \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\mathcal{R}^+(\alpha) >_* \mathcal{R}^+(\beta)$  and, by Definition 6,  $ST(\beta) <_* ST(\alpha)$ . Therefore, by Principle 2,  $\sigma(\alpha) \gg \sigma(\beta)$  and strict monotonicity implies GP8.

If  $\mathcal{R}_*^-(\alpha) = \mathcal{R}_*^-(\beta)$ ,  $\mathcal{R}_*^+(\alpha) = \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then, by Definition 6,  $ST(\alpha) <_* ST(\beta)$ . Therefore, by Principle 2,  $\sigma(\alpha) \ll \sigma(\beta)$  and monotonicity implies GP9.

If  $\mathcal{R}^-(\alpha) <_* \mathcal{R}^-(\beta)$ ,  $\mathcal{R}_*^+(\alpha) = \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then, by Definition 6,  $ST(\beta) <_* ST(\alpha)$ . Therefore, by Principle 2,  $\sigma(\beta) \ll \sigma(\alpha)$  and strict monotonicity implies GP10.

If  $\mathcal{R}_*^-(\alpha) = \mathcal{R}_*^-(\beta)$ ,  $\mathcal{R}_*^+(\alpha) >_* \mathcal{R}_*^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then, by Definition 6,  $ST(\beta) <_* ST(\alpha)$ . Therefore, by Principle 2,  $\sigma(\beta) \ll \sigma(\alpha)$  and strict monotonicity implies GP11.  $\square$

**Proposition 22.** The  $h$ -categoriser is strictly balanced and strictly monotonic for any choice of  $*$  and  $\ll$  (i.e. for  $\ll = <$ ).

**Proof.** For every argument  $\alpha$  in this case  $\mathcal{R}^+(\alpha) = \emptyset$ .

Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha) = 1$ , which is clearly satisfied by (1). Point 2 is equivalent to: if  $\mathcal{R}^-(\alpha) \neq \emptyset$  then  $\sigma(\alpha) < 1$  which is also satisfied by (1), given that if  $\mathcal{R}^-(\alpha) \neq \emptyset$  then  $\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) > 0$ . Finally, Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_* \emptyset$  cannot occur. Therefore, the  $h$ -categoriser is balanced.

Further, Point 4 becomes: if  $\sigma(\alpha) < 1$  then  $\mathcal{R}^-(\alpha) >_* \emptyset$ , i.e.  $\exists \beta \in \mathcal{R}^-(\alpha)$ , which is also satisfied by (1), given that if  $\sigma(\alpha) < 1$  then  $\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) > 0$  and, therefore,  $\exists \beta \in \mathcal{R}^-(\alpha)$ . Finally, Point 5 is irrelevant since  $\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) \geq 0$  and hence  $\sigma(\alpha) > \tau(\alpha) = 1$  cannot occur. Therefore, the  $h$ -categoriser is strictly balanced.

For every argument  $\alpha$  and  $\beta$  we have  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta) = \emptyset$ . Then, in this case strict monotonicity is equivalent to if  $\mathcal{R}^-(\alpha) <_* \mathcal{R}^-(\beta)$  then  $\sigma(\alpha) > \sigma(\beta)$ . Given that for every argument  $\sigma(\alpha) > 0$ , we have that  $\mathcal{R}^-(\beta) <_* \mathcal{R}^-(\alpha)$  implies  $\sum_{\delta \in \mathcal{R}^-(\beta)} \sigma(\delta) < \sum_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma)$  and then  $\sigma(\alpha) > \sigma(\beta)$  from (1). Therefore, the  $h$ -categoriser is strictly monotonic.  $\square$

**Proposition 23.** The inverse equational system is strictly balanced and strictly monotonic with  $*$  =  $\sigma \perp$  and  $\ll = <_{\perp}$ .

**Proof.** Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha) = 1$ , which is satisfied since the default value of 1 is assigned in absence of attackers. Point 2 is equivalent to: if  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$  which is satisfied by (2), given that if  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$  then  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > 0$  and then  $\prod_{\beta \in \mathcal{R}^-(\alpha)} (1 - \sigma(\beta)) < 1$ . Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \emptyset$  cannot occur. Further, Point 4 becomes: if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$ , i.e.  $\exists \beta \in \mathcal{R}^-(\alpha)$  with  $\sigma(\beta) > 0$ , which is also satisfied by (2), given that if  $\sigma(\alpha) < \tau(\alpha)$  then  $\prod_{\beta \in \mathcal{R}^-(\alpha)} (1 - \sigma(\beta)) < 1$  and, therefore,  $\exists \beta \in \mathcal{R}^-(\alpha)$  with  $\sigma(\beta) > 0$ . Finally, Point 5 is irrelevant since  $\sigma(\alpha) > \tau(\alpha) = 1$  cannot occur. Therefore, the inverse equational system is strictly balanced. Turning to monotonicity, for every argument  $\alpha$  and  $\beta$  we have  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta) = \emptyset$  and  $\tau(\alpha) = \tau(\beta) = 1$ . Clearly, if also  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \mathcal{R}^-(\beta)$  we get from (2) that  $\sigma(\alpha) = \sigma(\beta)$ .

In this case strict monotonicity is equivalent to if  $\mathcal{R}^-(\beta) <_{\sigma \perp} \mathcal{R}^-(\alpha)$  then  $\sigma(\alpha) <_{\perp} \sigma(\beta)$ . Now  $\mathcal{R}^-(\beta) <_{\sigma \perp} \mathcal{R}^-(\alpha)$  means that for every  $\gamma \in \mathcal{R}^-(\beta)$  such that  $\sigma(\gamma) > 0$  it exists  $\delta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\delta) \geq \sigma(\gamma)$ . Moreover it holds that either  $\exists \gamma \in \mathcal{R}^-(\beta)$ ,  $\exists \delta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\delta) > \sigma(\gamma) > 0$  or  $|\mathcal{R}_{\sigma \perp}^-(\beta)| < |\mathcal{R}_{\sigma \perp}^-(\alpha)|$ . From the above conditions it follows that if  $\exists \gamma \in \mathcal{R}^-(\beta)$  such that  $\sigma(\gamma) = 1$  then also  $\exists \delta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\delta) = 1$  and then from (2)  $\sigma(\alpha) = \sigma(\beta) = 0$ . If instead  $\nexists \gamma \in \mathcal{R}^-(\beta)$  such that  $\sigma(\gamma) = 1$  then  $\sigma(\beta) > 0$  and the conditions listed above imply that  $\prod_{\gamma \in \mathcal{R}^-(\beta)} (1 - \sigma(\gamma)) > \prod_{\delta \in \mathcal{R}^-(\alpha)} (1 - \sigma(\delta))$ , i.e.  $\sigma(\alpha) < \sigma(\beta)$ , which completes the proof.  $\square$

**Proposition 24.** The suspect equational system is strictly balanced with  $*$  =  $\sigma \perp$  and  $\ll = \leq$ .



**Proof.** We first consider the cases when (3) applies, i.e.  $\alpha \notin \mathcal{R}^-(\alpha)$ . Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha) = 1$ , which is satisfied since the default value of 1 is assigned in absence of attackers. Point 2 is equivalent to: if  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$  then  $\sigma(\alpha) \leq \tau(\alpha)$  which is satisfied by (3), given that  $\sigma(\alpha) \in [0, 1]$ . Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \emptyset$  cannot occur. Further, Point 4 becomes: if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$ , which is also satisfied by (3), given that if  $\sigma(\alpha) < \tau(\alpha)$  then  $\prod_{\beta \in \mathcal{R}^-(\alpha)} (1 - \sigma(\beta)) < 1$  and, therefore,  $\exists \beta \in \mathcal{R}^-(\alpha)$ . Finally, Point 5 is irrelevant since  $\sigma(\alpha) > \tau(\alpha) = 1$  cannot occur.

We then prove the statement for the cases when (4) applies, i.e.  $\alpha \in \mathcal{R}^-(\alpha)$ . Point 1 is irrelevant since  $\alpha \in \mathcal{R}^-(\alpha)$  so the condition  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \emptyset$  does not hold. Point 2 is equivalent to: if  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$  then  $\sigma(\alpha) \leq \tau(\alpha)$  which is satisfied by (4) since  $\sigma(\alpha) = 0 = \perp$  in this case. Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \emptyset$  cannot occur. Further, Point 4 becomes: if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$ , which is satisfied by (4), given that  $\alpha \in \mathcal{R}^-(\alpha)$  and  $\sigma(\alpha) = 0$  here. Finally, Point 5 is irrelevant since  $\sigma(\alpha) > \tau(\alpha) = 1$  cannot occur. Therefore, the suspect equational system is strictly balanced.  $\square$

**Proposition 25.** *The max equational system is strictly balanced and monotonic with  $*$  =  $\sigma \perp$  and with any  $\ll$ .*

**Proof.** Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha) = 1$ , which is satisfied since the default value of 1 is assigned in absence of attackers. Point 2 is equivalent to: if  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$  which is satisfied by (5), given that if  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$  then  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > 0$  and then  $1 - \max_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) < 1$ . Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \emptyset$  cannot occur. Further, Point 4 becomes: if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$ , i.e.  $\exists \beta \in \mathcal{R}^-(\alpha)$  with  $\sigma(\beta) > 0$ , which is also satisfied by (5), given that if  $\sigma(\alpha) < \tau(\alpha)$  then we get  $1 - \max_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) < 1$  and, therefore,  $\exists \beta \in \mathcal{R}^-(\alpha)$  with  $\sigma(\beta) > 0$ . Finally, Point 5 is irrelevant since  $\sigma(\alpha) > \tau(\alpha) = 1$  cannot occur. Therefore, the max equational system is strictly balanced.

Turning to monotonicity, for any arguments  $\alpha$  and  $\beta$  we have  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta) = \emptyset$  and  $\tau(\alpha) = \tau(\beta) = 1$ . Clearly if  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \mathcal{R}^-(\beta)$  we get from (5) that  $\sigma(\alpha) = \sigma(\beta)$  and it remains to check that if  $\mathcal{R}^-(\beta) \leq_{\sigma \perp} \mathcal{R}^-(\alpha)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ . Now  $\mathcal{R}^-(\beta) \leq_{\sigma \perp} \mathcal{R}^-(\alpha)$  implies that  $\max_{\gamma \in \mathcal{R}^-(\beta)} \sigma(\gamma) \leq \max_{\delta \in \mathcal{R}^-(\alpha)} \sigma(\delta)$  and then from (5)  $\sigma(\alpha) \leq \sigma(\beta)$ , as desired.  $\square$

**Proposition 26.** *The GTAA semantics is strictly balanced with  $*$  =  $\sigma \perp$  and  $\ll = <$ .*

**Proof.** In this case, for every argument  $\alpha$ ,  $\mathcal{R}^+(\alpha) = \emptyset$ . Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \emptyset$  then  $\sigma(\alpha) = 1$ . This is satisfied by the borderline conditions since if  $\alpha$  has no attackers, using this argument alone in  $P$  will give  $\rho(P, O) = 1$  in every case and therefore  $\sigma(\alpha) = 1$ . Point 2 then becomes: if  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$  then  $\sigma(\alpha) < 1$ , which is also satisfied given that  $\phi(P, O) < 1$  and the only way that  $\rho(P, O) = 1$  is if  $P \leftarrow O = \emptyset$ , which cannot hold since  $\alpha \in P$ . Finally, Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \emptyset$  cannot occur. Therefore, the GTAA semantics is balanced.

Further, Point 4 becomes: if  $\sigma(\alpha) < 1$  then  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$ , which is satisfied given that if  $\sigma(\alpha) < 1$  and therefore  $\phi(P, O) < 1$  for all proponent strategies, then there is no  $P$  where  $P \leftarrow O = \emptyset$  i.e.  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$ . Finally, Point 5 is irrelevant since  $\sigma(\alpha) \leq 1$ . Therefore, the GTAA semantics is strictly balanced.  $\square$

**Proposition 27.** *LocMax is balanced and monotonic with  $*$  =  $\sigma \perp$  and  $\ll = <_0$ .*

**Proof.** Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) = 0$ , which is clearly satisfied by (8). As to point 2, from  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \mathcal{R}^+(\alpha)$  it follows that  $\max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) \leq \max_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma)$ , hence  $\sigma(\alpha) \leq 0 < \top = 1$ , hence  $\sigma(\alpha) <_0 \tau(\alpha)$ . Similarly, as to point 3, from  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \mathcal{R}^+(\alpha)$  it follows that  $\max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) \geq \max_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma)$ , hence  $\sigma(\alpha) \geq_0 \tau(\alpha) = -1$  and hence  $\sigma(\alpha) >_0 \tau(\alpha)$ . Therefore, LocMax semantics is balanced.

Given two arguments  $\alpha$  and  $\beta$  clearly if  $ST(\alpha) \simeq_{\sigma \perp} ST(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  from (8). If  $ST(\alpha) \leq_{\sigma \perp} ST(\beta)$  we have that  $\max_{\gamma \in \mathcal{R}^+(\alpha)} \sigma(\gamma) \leq \max_{\gamma \in \mathcal{R}^+(\beta)} \sigma(\gamma)$  and  $\max_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma) \geq \max_{\gamma \in \mathcal{R}^-(\beta)} \sigma(\gamma)$ . From equation (8) it follows that  $\sigma(\alpha) \leq \sigma(\beta)$ . Therefore, LocMax semantics is monotonic.  $\square$

**Proposition 28.** *LocSum is balanced and strictly monotonic with  $*$  =  $\sigma \perp$  and for any choice of  $\ll$  (i.e. with  $\ll = <$ ).*

**Proof.** Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) = 0$ , which is clearly satisfied by (9) since in this case  $h(\mathcal{R}^-(\alpha)) = h(\mathcal{R}^+(\alpha))$ . As to Point 2, from  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \mathcal{R}^+(\alpha)$  it follows that  $h(\mathcal{R}^-(\alpha)) > h(\mathcal{R}^+(\alpha))$ , hence  $\sigma(\alpha) < 0 = \tau(\alpha) < \top = 1$ . Similarly, as to Point 3, from  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \mathcal{R}^+(\alpha)$  it follows that  $h(\mathcal{R}^-(\alpha)) < h(\mathcal{R}^+(\alpha))$ , hence  $\sigma(\alpha) > 0 = \tau(\alpha) > \perp = -1$ . Therefore, LocSum semantics is balanced.

Given two arguments  $\alpha$  and  $\beta$  clearly if  $ST(\alpha) \simeq_{\sigma \perp} ST(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  from (9). If  $ST(\alpha) <_{\sigma \perp} ST(\beta)$  it follows that  $h(\mathcal{R}^+(\alpha)) \leq h(\mathcal{R}^+(\beta))$  and  $h(\mathcal{R}^-(\alpha)) \geq h(\mathcal{R}^-(\beta))$  with at least one of the above inequalities being strict and from (9) it follows that  $\sigma(\alpha) < \sigma(\beta)$ . Therefore, LocSum semantics is strictly monotonic.  $\square$

**Proposition 29.** *A well behaved  $S$  – model is monotonic with  $*$  =  $\sigma \perp$ . If the monotonicity properties of  $\wedge$ ,  $\vee$ , and  $\neg$  are strict, then a well behaved  $S$  – model is strictly balanced and strictly monotonic with  $*$  =  $\sigma \perp$  and for any choice of  $\ll$  (i.e. for  $\ll = <$ ).*

**Proof.** For every argument  $\alpha$  in this case  $\mathcal{R}^+(\alpha) = \emptyset$ .

Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ , which is satisfied by (10) given that  $\gamma\emptyset = \perp$  (see Proposition 11 in [10]). Point 2 is equivalent to: if  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$ . This is also satisfied by (10) if the monotonicity properties are strict: if  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$  then  $x = \gamma\{\sigma(\beta) \mid \beta \in \mathcal{R}^-(\alpha)\} > \perp$ , then  $\neg x < \top$  and  $\tau(\alpha) \wedge \neg x < \tau(\alpha)$ . Finally, Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma\perp} \emptyset$  cannot occur. Therefore, the  $S$ -model is balanced.

Further, Point 4 becomes: if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$ , i.e.  $\exists\beta \in \mathcal{R}^-(\alpha) : \sigma(\beta) > \perp$ , which is also satisfied by (10), given that if  $\sigma(\alpha) < \tau(\alpha)$  then  $\gamma\{\sigma(\beta) \mid \beta \in \mathcal{R}^-(\alpha)\} > \perp$  and, therefore,  $\exists\beta \in \mathcal{R}^-(\alpha) : \sigma(\beta) > \perp$ . Finally, Point 5 is irrelevant since  $\neg \gamma\{\sigma(\beta) \mid \beta \in \mathcal{R}^-(\alpha)\} \leq \top$ . Therefore, the  $S$ -model is strictly balanced.

Monotonicity follows directly from the definition of  $S$ -model.  $\square$

**Proposition 30.** The weighted max-based semantics is strictly balanced and monotonic with  $* = \sigma\perp$  and for any choice of  $\ll$  (i.e. with  $\ll = <$ ).

**Proof.** For every argument  $\alpha$  in this case  $\mathcal{R}^+(\alpha) = \emptyset$ . Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) =_{\sigma\perp} \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ , which is satisfied by (11) since  $\max_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) = 0$ . Point 2 then becomes: if  $\exists\beta \in \mathcal{R}^-(\alpha) : \sigma(\beta) > 0$  (thereby making  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$ ) then  $\sigma(\alpha) < \tau(\alpha)$ , which is also satisfied by (11), given that if  $\exists\beta \in \mathcal{R}^-(\alpha) : \sigma(\beta) > 0$  then  $\max_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) > 0$ . Finally, Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma\perp} \emptyset$  cannot occur. Therefore, weighted max-based semantics is balanced.

Further, Point 4 becomes: if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$ , i.e.  $\exists\beta \in \mathcal{R}^-(\alpha) : \sigma(\beta) > 0$ , which is also satisfied by (11), given that if  $\sigma(\alpha) < \tau(\alpha)$  then  $\max_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) > 0$ . Finally, Point 5 is irrelevant since  $\max_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) \geq 0$ . Therefore, weighted max-based semantics is strictly balanced.

Given two arguments  $\alpha$  and  $\beta$  clearly if  $\mathcal{ST}(\alpha) \simeq_{\sigma\perp} \mathcal{ST}(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  from (11). We need now to prove that if  $\mathcal{ST}(\alpha) <_{\sigma\perp} \mathcal{ST}(\beta)$ , meaning  $\tau(\alpha) \leq \tau(\beta)$  and  $\mathcal{R}^-(\alpha) \geq_{\sigma\perp} \mathcal{R}^-(\beta)$ , then  $\sigma(\alpha) \leq \sigma(\beta)$ . Since  $\mathcal{R}^-(\alpha) \geq_{\sigma\perp} \mathcal{R}^-(\beta)$ ,  $\max_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma) \geq \max_{\delta \in \mathcal{R}^-(\beta)} \sigma(\delta)$  as otherwise there could be no correspondence from  $\mathcal{R}^-(\alpha)$  to  $\mathcal{R}^-(\beta)$  such that  $\forall \gamma \in \mathcal{R}^-(\alpha), \sigma(f(\gamma)) \leq \sigma(\gamma)$ . Therefore,  $\frac{\tau(\alpha)}{1 + \max_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma)} \leq \frac{\tau(\beta)}{1 + \max_{\delta \in \mathcal{R}^-(\beta)} \sigma(\delta)}$  and  $\sigma(\alpha) \leq \sigma(\beta)$ . Therefore, weighted max-based semantics is monotonic.  $\square$

**Proposition 31.** The weighted card-based semantics is strictly balanced and strictly monotonic for  $* = \sigma\perp$  and any  $\ll$ .

**Proof.** For every argument  $\alpha$  in this case  $\mathcal{R}^+(\alpha) = \emptyset$ . Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) =_{\sigma\perp} \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ , which is satisfied by (12) since its denominator is equal to 1 in this case. Point 2 then becomes: if  $\exists\beta \in \mathcal{R}^-(\alpha)$  (thereby making  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$ ) then  $\sigma(\alpha) < \tau(\alpha)$ , which is also satisfied by (12), since its denominator is greater than 1 if  $|\mathcal{R}^-(\alpha)| > 0$ . Finally, Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma\perp} \emptyset$  cannot occur. Therefore, weighted card-based semantics is balanced.

Further, Point 4 becomes: if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$ , i.e.  $\exists\beta \in \mathcal{R}^-(\alpha)$ , which is also satisfied by (12), given that if  $\sigma(\alpha) < \tau(\alpha)$  then  $|\mathcal{R}^-(\alpha)| > 0$  or  $\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) > 0$ . Finally, Point 5 is irrelevant since the denominator in (12) is at least 1. Therefore, weighted card-based semantics is strictly balanced.

Given two arguments  $\alpha$  and  $\beta$  clearly if  $\mathcal{ST}(\alpha) \simeq_{\sigma\perp} \mathcal{ST}(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  from (12). We need now to prove that if  $\mathcal{ST}(\alpha) <_{\sigma\perp} \mathcal{ST}(\beta)$ , meaning  $\tau(\alpha) \geq \tau(\beta)$  and  $\mathcal{R}^-(\alpha) \leq_{\sigma\perp} \mathcal{R}^-(\beta)$ , then  $\sigma(\alpha) \geq \sigma(\beta)$ . We let  $x_a = |\mathcal{R}^-(\alpha)|$ ,  $x_b = |\mathcal{R}^-(\beta)|$ ,  $y_a = \sum_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma)$  and  $y_b = \sum_{\delta \in \mathcal{R}^-(\beta)} \sigma(\delta)$ . Now we know that  $x_a \leq x_b$  and  $y_a \leq y_b$  since  $\mathcal{R}^-(\alpha) \leq_{\sigma\perp} \mathcal{R}^-(\beta)$  implies  $|\mathcal{R}^-(\alpha)| \leq |\mathcal{R}^-(\beta)|$  and  $\sum_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma) \leq \sum_{\delta \in \mathcal{R}^-(\beta)} \sigma(\delta)$ , resp. Now if  $x_a = 0$  we get  $\sigma(\alpha) = 0$  and then  $\sigma(\alpha) \leq \sigma(\beta)$ .

We can then assume  $0 < x_a$  in the following. Then,  $\mathcal{R}^-(\alpha) \leq_{\sigma\perp} \mathcal{R}^-(\beta)$  implies  $x_a + \frac{y_a}{x_a} \leq x_b + \frac{y_b}{x_b}$  iff  $\frac{x_a^2 + y_a}{x_a} \leq \frac{x_b^2 + y_b}{x_b}$ , which becomes  $x_a^2 x_b + x_b y_a \leq x_a x_b^2 + x_a y_b$  and then  $x_a x_b (x_a - x_b) \leq x_a y_b - x_b y_a$ . Now since  $x_a, x_b, y_a, y_b \geq 0$  and  $y_a \leq y_b$  it holds that  $x_a y_a - x_b y_a \leq x_a y_b - x_b y_a$ , thus if  $x_a x_b (x_a - x_b) \leq x_a y_a - x_b y_a$  then  $x_a x_b (x_a - x_b) \leq x_a y_b - x_b y_a$ . To prove this we require  $x_a x_b (x_a - x_b) \leq y_a (x_a - x_b)$  and, given that  $(x_a - x_b) \leq 0$ ,  $y_a \geq 0$  and  $x_a x_b \geq 0$ , this holds iff  $x_a x_b \geq y_a$ , which holds because either  $1 \leq x_a \leq x_b$  and  $y_a \leq x_a$  or  $0 = x_a \leq x_b$  and  $y_a = 0$ . As to strict monotonicity, if  $\mathcal{ST}(\beta) <_{\sigma\perp} \mathcal{ST}(\alpha)$  we get that  $\tau(\alpha) > \tau(\beta)$  or  $x_a < x_b$  or  $y_a < y_b$ , each of these strict conditions individually and any of their combinations leading to derive a strict inequality  $\sigma(\alpha) < \sigma(\beta)$  using the same line of reasoning as above.  $\square$

**Proposition 32.** The weighted h-categoriser is strictly balanced and strictly monotonic with  $* = \sigma\perp$  and for any  $\ll$ .

**Proof.** For every argument  $\alpha$  in this case  $\mathcal{R}^+(\alpha) = \emptyset$ .

Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ , which is clearly satisfied by (13). Point 2 is equivalent to: if  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$  which is also satisfied by (13), given that if  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$  then  $\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) > 0$ . Finally, Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma\perp} \emptyset$  cannot occur. Therefore, the weighted h-categoriser is balanced.

Further, Point 4 becomes: if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) >_{\sigma\perp} \emptyset$ , i.e.  $\exists\beta \in \mathcal{R}^-(\alpha)$  with  $\sigma(\beta) > 0$ , which is also satisfied by (13), given that if  $\sigma(\alpha) < \tau(\alpha)$  then  $\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) > 0$  and, therefore,  $\exists\beta \in \mathcal{R}^-(\alpha)$  with  $\sigma(\beta) > 0$ . Finally, Point 5 is

irrelevant since  $\sum_{\beta \in \mathcal{R}^-(\alpha)} \sigma(\beta) \geq 0$  and hence  $\sigma(\alpha) > \tau(\alpha)$  cannot occur. Therefore, the weighted h-categoriser is strictly balanced.

For every argument  $\alpha$  and  $\beta$  we have  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta) = \emptyset$ . Then, in this case strict monotonicity is equivalent to (i) if  $\tau(\alpha) = \tau(\beta)$  and  $\mathcal{R}^-(\beta) =_{\sigma \perp} \mathcal{R}^-(\alpha)$  then  $\sigma(\alpha) = \sigma(\beta)$ , (ii) if  $\tau(\alpha) > \tau(\beta)$  and  $\mathcal{R}^-(\beta) \leq_{\sigma \perp} \mathcal{R}^-(\alpha)$  then  $\sigma(\alpha) > \sigma(\beta)$ , and (iii) if  $\tau(\alpha) \leq \tau(\beta)$  and  $\mathcal{R}^-(\beta) <_{\sigma \perp} \mathcal{R}^-(\alpha)$  then  $\sigma(\alpha) > \sigma(\beta)$ . Condition (i) is clearly satisfied by (13). As to conditions (ii) and (iii), given that  $\mathcal{R}^-(\beta) \leq_{\sigma \perp} \mathcal{R}^-(\alpha)$  implies  $\sum_{\gamma \in \mathcal{R}^-(\beta)} \sigma(\gamma) \leq \sum_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma)$  and  $\mathcal{R}^-(\beta) <_{\sigma \perp} \mathcal{R}^-(\alpha)$  implies  $\sum_{\gamma \in \mathcal{R}^-(\beta)} \sigma(\gamma) < \sum_{\gamma \in \mathcal{R}^-(\alpha)} \sigma(\gamma)$ , they are also satisfied by (13). Therefore, the weighted h-categoriser is strictly monotonic.  $\square$

**Proposition 33.** *The top-based semantics is strictly balanced and monotonic with  $* = \sigma \perp$  and  $\ll = <_{\top}$ .*

**Proof.** For every argument  $\alpha$  in this case  $\mathcal{R}^-(\alpha) = \emptyset$ .

Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ , which is clearly satisfied by (14). Point 2 is irrelevant since the case  $\emptyset >_{\sigma \perp} \mathcal{R}^+(\alpha)$  cannot occur. Finally, Point 3 is equivalent to if  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) >_{\top} \tau(\alpha)$ . If  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) > 0$  and in Equation (14) we have that the term  $(1 - \tau(\alpha)) \max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) > 0$  or  $\tau(\alpha) = 1$ . It follows that either  $\sigma(\alpha) = \tau(\alpha) = 1 = \top$  or  $\sigma(\alpha) > \tau(\alpha)$  i.e.  $\sigma(\alpha) >_{\top} \tau(\alpha)$ . Therefore, top-based semantics is balanced.

Further, Point 4 is irrelevant since  $\max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) \geq 0$ . Finally, Point 5 becomes: if  $\sigma(\alpha) > \tau(\alpha)$  then  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$ , i.e.  $\exists \beta \in \mathcal{R}^+(\alpha) : \sigma(\beta) > 0$ , which is also satisfied by (14), given that if  $\sigma(\alpha) > \tau(\alpha)$  then  $\max_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) > 0$ . Therefore, top-based semantics is strictly balanced.

Given two arguments  $\alpha$  and  $\beta$  clearly if  $\mathcal{ST}(\alpha) \simeq_{\sigma \perp} \mathcal{ST}(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  from (14). If  $\mathcal{ST}(\alpha) \leq_{\sigma \perp} \mathcal{ST}(\beta)$  we have that  $\tau(\alpha) \leq \tau(\beta)$  and  $\max_{\gamma \in \mathcal{R}^+(\alpha)} \sigma(\gamma) \leq \max_{\gamma \in \mathcal{R}^+(\beta)} \sigma(\gamma)$ . The fact that  $\sigma(\alpha) \leq \sigma(\beta)$  follows then from the fact that for  $x, y$  in  $[0, 1]$  the function  $f(x, y) = x + (1 - x)y$  is a T-conorm (called probabilistic sum) and is monotonic with respect to both arguments, i.e.  $f(x, y) \leq f(x', y')$  if  $x \leq x'$  and  $y \leq y'$ . Therefore, top-based semantics is monotonic.  $\square$

**Proposition 34.** *The reward-based semantics is strictly balanced and strictly monotonic with  $* = \sigma \perp$  and  $\ll = <_{\top}$ .*

**Proof.** For every argument  $\alpha$  in this case  $\mathcal{R}^-(\alpha) = \emptyset$ .

Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ , which is clearly satisfied by (15). Point 2 is irrelevant since the case  $\emptyset >_{\sigma \perp} \mathcal{R}^+(\alpha)$  cannot occur. Finally, Point 3 is equivalent to if  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) >_{\top} \tau(\alpha)$ . If  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\mathcal{R}_F^+(\alpha) \neq \emptyset$  and in Equation (15) we have that the term  $(1 - \tau(\alpha)) \left( \sum_{j=1}^{n(\alpha)-1} \frac{1}{2^j} + \frac{m(\alpha)}{2^{n(\alpha)}} \right) > 0$  or  $\tau(\alpha) = 1$ . It follows that either  $\sigma(\alpha) = \tau(\alpha) = 1 = \top$  or  $\sigma(\alpha) > \tau(\alpha)$  i.e.  $\sigma(\alpha) >_{\top} \tau(\alpha)$ . Therefore, reward-based semantics is balanced.

Further, Point 4 is irrelevant since  $\sigma(\alpha) \geq \tau(\alpha)$ . Finally, Point 5 becomes: if  $\sigma(\alpha) > \tau(\alpha)$  then  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$ , i.e.  $\exists \beta \in \mathcal{R}^+(\alpha) : \sigma(\beta) > 0$ , which is also satisfied by (15), given that if  $\sigma(\alpha) > \tau(\alpha)$  then  $\sum_{j=1}^{n(\alpha)-1} \frac{1}{2^j} + \frac{m(\alpha)}{2^{n(\alpha)}} > 0$ . Therefore, reward-based semantics is strictly balanced.

Given two arguments  $\alpha$  and  $\beta$  clearly if  $\mathcal{ST}(\alpha) \simeq_{\sigma \perp} \mathcal{ST}(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  from (14). If  $\mathcal{ST}(\alpha) <_{\sigma \perp} \mathcal{ST}(\beta)$  it follows that  $\tau(\alpha) \leq \tau(\beta)$ ,  $|\mathcal{R}_F^+(\alpha)| \leq |\mathcal{R}_F^+(\beta)|$  and  $\sum_{\gamma \in \mathcal{R}_F^+(\alpha)} \sigma(\gamma) \leq \sum_{\gamma \in \mathcal{R}_F^+(\beta)} \sigma(\gamma)$  with at least one of the above inequalities being strict. This in turn entails that  $\sum_{j=1}^{n(\alpha)-1} \frac{1}{2^j} + \frac{m(\alpha)}{2^{n(\alpha)}} \leq \sum_{j=1}^{n(\beta)-1} \frac{1}{2^j} + \frac{m(\beta)}{2^{n(\beta)}}$  and either this inequality is strict or  $\tau(\alpha) < \tau(\beta)$ . Then,  $\sigma(\alpha) <_{\top} \sigma(\beta)$  follows from the fact that for  $x, y$  in  $[0, 1]$  for the function  $f(x, y) = x + (1 - x)y$  it holds that if  $x \leq x'$  and  $y \leq y'$  and at least one of the inequalities is strict, then  $f(x, y) < f(x', y')$  unless  $x = 1$  or  $y = 1$ , in which case obviously  $f(x, y) = f(x', y') = 1 = \top$ . Therefore, reward-based semantics is strictly monotonic.  $\square$

**Proposition 35.** *The aggregation-based semantics is strictly balanced and strictly monotonic with  $* = \sigma \perp$  and  $\ll = <_{\top}$ .*

**Proof.** For every argument  $\alpha$  in this case  $\mathcal{R}^-(\alpha) = \emptyset$ .

Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) = \tau(\alpha)$ , which is clearly satisfied by (16). Point 2 is irrelevant since the case  $\emptyset >_{\sigma \perp} \mathcal{R}^+(\alpha)$  cannot occur. Finally, Point 3 is equivalent to if  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) >_{\top} \tau(\alpha)$ . If  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) > 0$  and  $\frac{\sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta)}{1 + \sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta)} > 0$ . It follows that either  $\sigma(\alpha) = \tau(\alpha) = 1 = \top$  or  $\sigma(\alpha) > \tau(\alpha)$  i.e.  $\sigma(\alpha) >_{\top} \tau(\alpha)$ . Therefore, aggregation-based semantics is balanced.

Further, Point 4 is irrelevant since  $\sigma(\alpha) \geq \tau(\alpha)$ . Finally, Point 5 becomes: if  $\sigma(\alpha) > \tau(\alpha)$  then  $\emptyset <_{\sigma \perp} \mathcal{R}^+(\alpha)$ , i.e.  $\exists \beta \in \mathcal{R}^+(\alpha) : \sigma(\beta) > 0$ , which is also satisfied by (16), given that if  $\sigma(\alpha) > \tau(\alpha)$  then  $\sum_{\beta \in \mathcal{R}^+(\alpha)} \sigma(\beta) > 0$ . Therefore, aggregation-based semantics is strictly balanced.

Given two arguments  $\alpha$  and  $\beta$  clearly if  $\mathcal{ST}(\alpha) \simeq_{\sigma \perp} \mathcal{ST}(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  from (16). If  $\mathcal{ST}(\alpha) <_{\sigma \perp} \mathcal{ST}(\beta)$  it follows that  $\tau(\alpha) \leq \tau(\beta)$  and  $\sum_{\gamma \in \mathcal{R}^+(\alpha)} \sigma(\gamma) \leq \sum_{\gamma \in \mathcal{R}^+(\beta)} \sigma(\gamma)$  with at least one of the above inequalities being strict. This in turn entails that  $\frac{\sum_{\gamma \in \mathcal{R}^+(\alpha)} \sigma(\gamma)}{1 + \sum_{\gamma \in \mathcal{R}^+(\alpha)} \sigma(\gamma)} \leq \frac{\sum_{\gamma \in \mathcal{R}^+(\beta)} \sigma(\gamma)}{1 + \sum_{\gamma \in \mathcal{R}^+(\beta)} \sigma(\gamma)}$  and either this inequality is strict or  $\tau(\alpha) < \tau(\beta)$  (note in particular that the function  $\frac{x}{1+x}$  is strictly monotonic w.r.t.  $x$ ). Then, as above, the fact that  $\sigma(\alpha) <_{\top} \sigma(\beta)$  follows then from the fact that for

$x, y$  in  $[0, 1]$  for the function  $f(x, y) = x + (1 - x)y$  it holds that if  $x \leq x'$  and  $y \leq y'$  and at least one of the inequalities is strict, then  $f(x, y) < f(x', y')$  unless  $x = 1$  or  $y = 1$ , in which case obviously  $f(x, y) = f(x', y') = 1 = \top$ . Therefore, the aggregation-based semantics is strictly monotonic.  $\square$

**Proposition 36.** *The QuAD algorithm is monotonic with  $* = \sigma \perp$ .*

**Proof.** Let  $P$  be a sequence over  $(0, 1]$ ,  $Q$  be a sequence where all elements are 0, and  $R$  be the concatenation of  $P$  and  $Q$ . Then, for either  $\times = a$  or  $\times = s$ ,  $\mathcal{F}_\times(R) = \mathcal{F}_\times(P)$  since, from (25) and (26),  $f_a(v_0, 0) = v_0$  and  $f_s(v_0, 0) = v_0$ , therefore attackers or supporters with the bottom strength have no effect on an argument's strength.

Then, for two sets of arguments  $S, T \subseteq \mathcal{X}$ , let  $S'$  and  $T'$  be the corresponding sets with the 0 values removed. If  $S =_{\sigma \perp} T$ , (23) and (24) show that  $\mathcal{F}_a(S) = \mathcal{F}_a(T)$  and  $\mathcal{F}_s(S) = \mathcal{F}_s(T)$ . Alternatively, let us consider a situation where  $S >_{\sigma \perp} T$  and there exists an injective mapping from  $T'$  to  $S'$  such that  $\forall \alpha \in T', \sigma(f(\alpha)) \geq \sigma(\alpha)$ . Now it can be seen from (25) ((26)) that if  $v > 0$  then  $f_a(v_0, v) <_{\perp} v_0$  ( $f_s(v_0, v) >_{\top} v_0$ , resp.). Also, for  $v_0, v'_0, v, v' \in \mathbb{I}$ , where  $v_0 = v'_0$  and  $v = v'$  unless differently stated, if  $v_0 > v'_0$  or  $v > v'$  then  $f_a(v_0, v) <_{\perp} f_a(v'_0, v')$  ( $f_s(v_0, v) >_{\top} f_s(v'_0, v')$ , resp.). Therefore, if  $S >_{\sigma \perp} T$ , then  $\mathcal{F}_a(S) <_{\perp} \mathcal{F}_a(T)$  ( $\mathcal{F}_s(S) >_{\top} \mathcal{F}_s(T)$ , resp.). This is shown in Proposition 3.3 of [9].

In the following, we assume that  $nil \leq 0$  and  $nil < x$  for every  $x \neq 0$ . Given two arguments  $\alpha$  and  $\beta$ , we let  $v_0 = \tau(\alpha)$ ,  $v'_0 = \tau(\beta)$ ,  $v_a = \mathcal{F}_a(\tau(\alpha), \sigma(\mathcal{R}^-(\alpha)))$ ,  $v'_a = \mathcal{F}_a(\tau(\beta), \sigma(\mathcal{R}^-(\beta)))$ ,  $v_s = \mathcal{F}_s(\tau(\alpha), \sigma(\mathcal{R}^+(\alpha)))$  and  $v'_s = \mathcal{F}_s(\tau(\beta), \sigma(\mathcal{R}^+(\beta)))$ . Clearly if  $\mathcal{ST}(\alpha) \simeq_{\sigma \perp} \mathcal{ST}(\beta)$  then  $v_0 = v'_0$ ,  $v_a = v'_a$  and  $v_s = v'_s$ . Now, for both  $\alpha$  and  $\beta$ , if (18) holds  $\sigma(\alpha) = \sigma(\beta) = v_a = v'_a$ . If (19) holds  $\sigma(\alpha) = \sigma(\beta) = v_s = v'_s$ . If (20) holds  $\sigma(\alpha) = \sigma(\beta) = v_0 = v'_0$ . Finally, if (21) holds  $\sigma(\alpha) = \sigma(\beta) = \frac{v_a + v'_a}{2} = \frac{v_s + v'_s}{2}$  and therefore  $\sigma(\alpha) = \sigma(\beta)$  in all cases.

If  $\mathcal{ST}(\alpha) \leq_{\sigma \perp} \mathcal{ST}(\beta)$  then  $v_0 \leq v'_0$ ,  $v_a \leq v'_a$  and  $v_s \leq v'_s$ . Now if (18) holds for  $\alpha$ ,  $v_s = nil$  and so  $\sigma(\alpha) = v_a$ . If (18) holds for  $\beta$ , then  $\sigma(\beta) = v'_a$  and so  $\sigma(\alpha) \leq \sigma(\beta)$ . If (19) holds for  $\beta$ , then  $\sigma(\beta) = v'_s$ . We know that  $v'_s \geq v'_0 \geq v_0 \geq v_a$  and so  $\sigma(\alpha) \leq \sigma(\beta)$ . If (20) holds for  $\beta$ , then  $\sigma(\beta) = v'_0$  and again  $\sigma(\alpha) \leq \sigma(\beta)$ . If (21) holds for  $\beta$ , then  $\sigma(\beta) = \frac{v'_a + v'_s}{2}$  with  $v'_s > v'_a > v_a$  and so  $\sigma(\alpha) \leq \sigma(\beta)$ .

If (19) holds for  $\alpha$ ,  $v_a = nil$  and so  $\sigma(\alpha) = v_s$ . In this case, since  $\mathcal{R}^-(\beta) \leq_{\sigma \perp} \mathcal{R}^-(\alpha) =_{\sigma \perp} \emptyset$ ,  $v'_a = nil$  and since  $\mathcal{R}^+(\beta) \geq_{\sigma \perp} \mathcal{R}^+(\alpha) > \emptyset$ ,  $v'_s \neq nil$ . Therefore (19) holds for  $\beta$ , giving  $\sigma(\beta) = v'_s$  and so  $\sigma(\alpha) \leq \sigma(\beta)$ .

If (20) holds for  $\alpha$ ,  $v_a = v_s = nil$  and so  $\sigma(\alpha) = v_0$ . In this case, since  $\mathcal{R}^-(\beta) \leq_{\sigma \perp} \mathcal{R}^-(\alpha) =_{\sigma \perp} \emptyset$ ,  $v'_a = nil$ . If (19) holds for  $\beta$ , then  $\sigma(\beta) = v'_s$ . We know that  $v'_s \geq v'_0 \geq v_0$  and so  $\sigma(\alpha) \leq \sigma(\beta)$ . If (20) holds for  $\beta$ , then  $\sigma(\beta) = v'_0$  and again  $\sigma(\alpha) \leq \sigma(\beta)$ .

Finally, if (21) holds for  $\alpha$ ,  $v_a \neq nil$  and  $v_s \neq nil$  so  $\sigma(\alpha) = \frac{v_a + v_s}{2}$ . In this case, since  $\mathcal{R}^+(\beta) \geq_{\sigma \perp} \mathcal{R}^+(\alpha) > \emptyset$ ,  $v'_s \neq nil$ . If (19) holds for  $\beta$ , then  $\sigma(\beta) = v'_s$ . We know that  $v'_s \geq v_s > v_a$  and so  $\sigma(\alpha) \leq \sigma(\beta)$ . If (21) holds for  $\beta$ , then  $\sigma(\beta) = \frac{v'_a + v'_s}{2}$  with  $\frac{v'_a + v'_s}{2} \geq \frac{v_a + v_s}{2}$  and so  $\sigma(\alpha) \leq \sigma(\beta)$ .

This means that in all cases given by (18) to (21),  $\sigma(\alpha) \leq \sigma(\beta)$  and so the QuAD algorithm is Monotonic.  $\square$

**Proposition 37.** *The DF-QuAD algorithm is balanced and monotonic with  $* = \sigma \perp$  and  $\ll = \leq$ .*

**Proof.** Let  $P$  be a sequence over  $(0, 1]$ ,  $Q$  be a sequence where all elements are 0, and  $R$  be the concatenation of  $P$  and  $Q$ . Then,  $\mathcal{F}'(R) = \mathcal{F}'(P)$  since, from (34),  $f'(v_1, 0) = v_1$ , therefore attackers or supporters with the bottom strength have no effect on an argument's strength.

Then, for two sets of arguments  $S, T \subseteq \mathcal{X}$ , let  $S'$  and  $T'$  be the corresponding sets with the 0 values removed. If  $S =_{\sigma \perp} T$ , (30) to (33) show that  $\mathcal{F}'(S) = \mathcal{F}'(T)$ . Alternatively, let us consider a situation where  $S >_{\sigma \perp} T$  and there exists an injective mapping from  $T'$  to  $S'$  such that  $\forall \alpha \in T', \sigma(f(\alpha)) \geq \sigma(\alpha)$ . Then, we have one of three cases: i. the cardinality is the same but at least one mapped element has greater value; ii. the cardinality is not the same, the injective mapping involves elements all with the same value and there is at least one additional non-zero value; iii. the cardinality is not the same, at least one mapped element has greater value and there is at least one additional non-zero value. We show that in all cases,  $\mathcal{F}'(S) >_{\top} \mathcal{F}'(T)$ . For case i, it can be seen from (34) that if  $v_2 > 0$  then  $f'(v_1, v_2) >_{\top} v_1$ . Also, for  $v_1, v'_1, v_2, v'_2 \in \mathbb{I}$ , where  $v_1 = v'_1$  and  $v_2 = v'_2$  unless differently stated, if  $v_1 > v'_1$  or  $v_2 > v'_2$  then  $f'(v_1, v_2) >_{\top} f'(v'_1, v'_2)$ . Therefore, by (30) to (33),  $\mathcal{F}'(S) >_{\top} \mathcal{F}'(T)$ . For case ii, (33) and (34) show that if  $S \supsetneq T$ , if  $\exists v \in S \setminus T$  such that  $v > 0$ , then  $\mathcal{F}'(S) >_{\top} \mathcal{F}'(T)$ . Case iii is a combination of case i and case ii so, by the above, and therefore in all cases where  $S >_{\sigma \perp} T$ ,  $\mathcal{F}'(S) >_{\top} \mathcal{F}'(T)$ .

Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) = \tau(\alpha)$ , which is satisfied by (27) on the basis of (28) and (29) since in this case  $\mathcal{F}'(\sigma(\mathcal{R}^-(\alpha))) = \mathcal{F}'(\sigma(\mathcal{R}^+(\alpha)))$ . As to Point 2, from  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \mathcal{R}^+(\alpha)$  it follows that  $\mathcal{F}'(\sigma(\mathcal{R}^-(\alpha))) >_{\top} \mathcal{F}'(\sigma(\mathcal{R}^+(\alpha)))$ , hence, by (28),  $\sigma(\alpha) \leq \tau(\alpha)$ . Similarly, as to Point 3, from  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \mathcal{R}^+(\alpha)$  it follows that  $\mathcal{F}'(\sigma(\mathcal{R}^-(\alpha))) <_{\top} \mathcal{F}'(\sigma(\mathcal{R}^+(\alpha)))$ , hence, by (29),  $\sigma(\alpha) \geq \tau(\alpha)$ . Therefore, the DF-QuAD algorithm is balanced.

Given two arguments  $\alpha$  and  $\beta$ , we let  $v_0 = \tau(\alpha)$ ,  $v'_0 = \tau(\beta)$ ,  $v_a = \mathcal{F}'(\mathcal{R}^-(\alpha))$ ,  $v'_a = \mathcal{F}'(\mathcal{R}^-(\beta))$ ,  $v_s = \mathcal{F}'(\mathcal{R}^+(\alpha))$  and  $v'_s = \mathcal{F}'(\mathcal{R}^+(\beta))$ . Clearly, if  $\mathcal{ST}(\alpha) \simeq_{\sigma \perp} \mathcal{ST}(\beta)$  then  $v_0 = v'_0$ ,  $v_a = v'_a$  and  $v_s = v'_s$  therefore  $\sigma(\alpha) = \sigma(\beta)$  by (27).

If  $\mathcal{ST}(\alpha) \leq_{\sigma \perp} \mathcal{ST}(\beta)$  then  $v_0 \leq v'_0$ ,  $v_a \geq v'_a$  and  $v_s \leq v'_s$ . Now, let (28) hold for  $\alpha$  such that  $v_a \geq v_s$ . If (28) also holds for  $\beta$  then  $v'_a \geq v'_s$  and therefore  $|v_s - v_a| \geq |v'_s - v'_a|$ . Given that  $v_0 \leq v'_0$ , it can then be seen that  $v_0 - v_0 \cdot |v_s - v_a| \leq v'_0 - v'_0 \cdot |v'_s - v'_a|$  and  $\sigma(\alpha) \leq \sigma(\beta)$ . Otherwise, let (29) hold for  $\beta$  such that  $v'_a < v'_s$ . By inspection,  $c(v_0, v_a, v_s) \leq v_0 \leq v'_0 \leq c(v'_0, v'_a, v'_s)$  and so  $\sigma(\alpha) \leq \sigma(\beta)$ .

Now, let (29) hold for  $\alpha$  such that  $v_a < v_s$ . In this case it is not possible for (28) to hold for  $\beta$  because this would require  $v'_a \geq v'_s$ , which is incompatible with  $v_a < v_s$ ,  $v_a \geq v'_a$  and  $v_s \leq v'_s$ . Otherwise, let (29) also hold for  $\beta$  such that  $v'_a < v'_s$ . Then, given that  $|v_s - v_a| \leq |v'_s - v'_a|$  and  $v_0 \leq v'_0$ , it can then be seen that  $v_0 + (1 - v_0) \cdot |v_s - v_a| \leq v'_0 + (1 - v'_0) \cdot |v'_s - v'_a|$  and  $\sigma(\alpha) \leq \sigma(\beta)$ .

This means that in all cases given by (28) to (29),  $\sigma(\alpha) \leq \sigma(\beta)$  and so the DF-QuAD algorithm is Monotonic.  $\square$

**Proposition 38.** *The restricted Euler-based semantics is balanced and strictly monotonic with  $*$  =  $\sigma \perp$  and  $\ll = <_{\perp, \top}$ .*

**Proof.** Firstly we rearrange (35) to obtain  $(1 - \sigma(\alpha))(1 + e^E \cdot \tau(\alpha)) = (1 - \tau(\alpha))(1 + \tau(\alpha))$ . Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \mathcal{R}^+(\alpha)$  then  $\sigma(\alpha) = \tau(\alpha)$ . If the former is the case then by (36),  $E = 0$  and  $e^E = 1$  so, by (35),  $(1 - \sigma(\alpha))(1 + \tau(\alpha)) = (1 - \tau(\alpha))(1 + \tau(\alpha))$ , hence  $\sigma(\alpha) = \tau(\alpha)$ . As to point 2, from  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \mathcal{R}^+(\alpha)$  it follows from (36) that  $E < 0$  and  $e^E < 1$ , hence  $(1 - \sigma(\alpha))(1 + e^E \cdot \tau(\alpha)) = (1 - \tau(\alpha))(1 + \tau(\alpha))$ , where  $e^E \cdot \tau(\alpha) < \tau(\alpha)$ . Here, unless  $\tau(\alpha) = 0$  or  $\tau(\alpha) = 1$ , it must be the case that  $\sigma(\alpha) < \tau(\alpha)$ . Similarly, as to point 3, from  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \mathcal{R}^+(\alpha)$  it follows from (36) that  $E > 0$  and  $e^E > 1$ , hence  $(1 - \sigma(\alpha))(1 + e^E \cdot \tau(\alpha)) = (1 - \tau(\alpha))(1 + \tau(\alpha))$ , where  $e^E \cdot \tau(\alpha) > \tau(\alpha)$ . Here, unless  $\tau(\alpha) = 0$  or  $\tau(\alpha) = 1$ , it must be the case that  $\sigma(\alpha) > \tau(\alpha)$ . Therefore, restricted Euler-based semantics is balanced.

Given two arguments  $\alpha$  and  $\beta$  clearly if  $\mathcal{ST}(\alpha) \simeq_{\sigma \perp} \mathcal{ST}(\beta)$  then  $\sigma(\alpha) = \sigma(\beta)$  from (35) and (36) since  $E_\alpha = E_\beta$  and  $\tau(\alpha) = \tau(\beta)$ . If  $\mathcal{ST}(\alpha) <_{\sigma \perp} \mathcal{ST}(\beta)$  then  $\tau(\alpha) < \tau(\beta)$ ,  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) <_{\sigma \perp} \mathcal{R}^+(\beta)$  or a combination of these applies. If  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \mathcal{R}^-(\beta)$  or  $\mathcal{R}^+(\alpha) <_{\sigma \perp} \mathcal{R}^+(\beta)$  (or both) apply then  $E_\alpha < E_\beta$  and  $e^{E_\alpha} < e^{E_\beta}$ . By inspection this shows that  $\sigma(\alpha) < \sigma(\beta)$  unless  $\tau(\alpha) = \tau(\beta) = \sigma(\alpha) = \sigma(\beta) = 0$  or  $\tau(\alpha) = \tau(\beta) = \sigma(\alpha) = \sigma(\beta) = 1$ . If  $\tau(\alpha) < \tau(\beta)$ , then  $\tau(\beta) \neq 0$ . If  $\tau(\beta) = 1$  then  $\sigma(\beta) = 1$  while  $\tau(\alpha) < 1$  so, by inspection,  $\sigma(\alpha) < 1$  thus  $\sigma(\alpha) < \sigma(\beta)$ . If  $\tau(\beta) < 1$  then we must prove that  $1 - \frac{1-x^2}{1+xz} < 1 - \frac{1-y^2}{1+yz}$  where  $x < y$ . Cancelling and rearranging we obtain  $(1 - x^2)(1 + yz) > (1 - y^2)(1 + xz)$  then by inspection it can be seen that  $(1 - x^2) > (1 - y^2)$  and  $(1 + yz) > (1 + xz)$  so  $\sigma(\alpha) < \sigma(\beta)$ . Therefore, if  $\mathcal{ST}(\alpha) <_{\sigma \perp} \mathcal{ST}(\beta)$ , then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = 0$  or  $\sigma(\alpha) = \sigma(\beta) = 1$  and therefore the restricted Euler-based semantics is strictly monotonic.  $\square$

**Proposition 39.** *Every complete labelling is strictly balanced and monotonic with  $*$  =  $\sigma \perp$  and with any  $\ll$ .*

**Proof.** Recall that in argumentation frameworks, for every argument  $\alpha$  it holds  $\mathcal{R}^+(\alpha) = \emptyset$  and let us consider balance first. Then Point 1 of Principle 1 is equivalent to: if  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \emptyset$  (i.e.  $\forall \beta \in \mathcal{R}^-(\alpha) L(\beta) = \text{out}$ ) then  $\sigma(\alpha) = L(\alpha) = \text{in}$ , which is satisfied. Point 2 is equivalent to: if  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \emptyset$  (i.e.  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $L(\beta) \neq \text{out}$ ) then  $L(\alpha) \neq \text{in}$ , which is satisfied. Point 3 is irrelevant since the case  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \emptyset$  cannot occur. Point 4 corresponds to: if  $L(\alpha) \neq \text{in}$  then  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $L(\beta) \neq \text{out}$ , which, again is clearly satisfied. Finally Point 5 is irrelevant since  $L(\alpha) > \text{in}$  cannot occur. In summary, any complete labelling is strictly balanced.

As to monotonicity, if  $\mathcal{ST}(\alpha) \simeq_{\sigma \perp} \mathcal{ST}(\beta)$ , i.e. in particular  $\mathcal{R}^-(\alpha) =_{\sigma \perp} \mathcal{R}^-(\beta)$ , we get that  $L(\alpha) = L(\beta)$ .

Moreover, if  $\mathcal{ST}(\alpha) <_{\sigma \perp} \mathcal{ST}(\beta)$ , we get that it must be the case that  $\mathcal{R}^-(\alpha) >_{\sigma \perp} \mathcal{R}^-(\beta)$ , which means that there exists an injective mapping  $f$  from  $\mathcal{R}^-(\beta)$  to  $\mathcal{R}^-(\alpha)$  such that  $\forall \gamma \in \mathcal{R}^-(\beta)$ ,  $L(f(\gamma)) \geq L(\gamma)$  and either  $|\mathcal{R}^-(\alpha)| > |\mathcal{R}^-(\beta)|$  or  $\exists \delta \in \mathcal{R}^-(\alpha) : L(f(\delta)) > L(\delta)$ . From these conditions we get that:

- if  $L(\beta) = \text{in}$  (i.e.  $\mathcal{R}^-(\beta) = \emptyset$ ) then it must be the case that  $L(\alpha) \neq \text{in}$  since it must be the case that  $\mathcal{R}^-(\alpha) \neq \emptyset$ , i.e.  $\alpha$  has at least an attacker labelled in or und;
- if  $L(\beta) = \text{out}$  then it must be the case that  $L(\alpha) = \text{out}$ , since also  $\alpha$  must have at least an attacker labelled in;
- if  $L(\beta) = \text{und}$  then it must be the case that  $L(\alpha) = \text{und}$  or  $L(\alpha) = \text{out}$ , since also  $\alpha$  must have at least an attacker labelled und or possibly in.

It follows that  $L(\alpha) \leq L(\beta)$  and complete semantics satisfies monotonicity.  $\square$

## Appendix B. Additional hyphen properties

Here, we give novel properties corresponding to the hyphens in Table 2 for GP2 to GP11 and all of those in Table 3. Moreover we provide more general versions for the properties which are listed as implied in Table 2. To the best of our knowledge, these properties have not already been identified in the previous literature. While the list in this appendix is somehow arid it is meant to provide an exhaustive reference for future use. Also, this long list provides a sort of visual evidence supporting the utility of more synthetic principles as proposed in this paper. Some short comments on specific properties are given. For a more extensive discussion focused on the properties belonging to GP2, GP3, and GP9 the reader may refer to [21].

The hyphens in Table 2 in the rows for GP2 correspond to the following novel properties:

GP2 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \neq \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) \neq \emptyset$  and  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$ .



GP2 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \neq \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \perp$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) \neq \emptyset$  and  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \perp$ .

GP2 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \neq \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \top$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) \neq \emptyset$  and  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) < \tau(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \top$ .

The two latter properties may seem somewhat counter-intuitive in allowing arguments with top base score to have top strength even when they are attacked. However, here and later, this behaviour may be useful when arguments with a top base score need to have a special “tautological” status.

GP2 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \neq \emptyset$  then  $\sigma(\alpha) \leq \tau(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) \neq \emptyset$  and  $\mathcal{R}^+(\alpha) = \emptyset$  then  $\sigma(\alpha) \leq \tau(\alpha)$ .

GP2 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\sigma(\alpha) < \tau(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\nexists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\sigma(\alpha) < \tau(\alpha)$ .

GP2 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\sigma(\alpha) < \tau(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \perp$ .<sup>15</sup>
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\nexists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\sigma(\alpha) < \tau(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \perp$ .

GP2 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\sigma(\alpha) < \tau(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \top$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\nexists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\sigma(\alpha) < \tau(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \top$ .

Again, the latter two properties may be useful if arguments with top base score need to have special status.

GP2 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\sigma(\alpha) \leq \tau(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\nexists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\sigma(\alpha) \leq \tau(\alpha)$ .

The hyphens in Table 2 in the rows for GP3 correspond to the following properties:

GP3 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) < \sigma(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) < \sigma(\alpha)$ .

GP3 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) < \sigma(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \perp$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) < \sigma(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \perp$ .

The two latter properties may seem somewhat counter-intuitive in allowing arguments with bottom base score to have bottom strength even when they are supported. However, here and later, this behaviour may be useful when arguments with a bottom base score have a special “fallacious” status.

GP3 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) < \sigma(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \top$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) < \sigma(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \top$ .

<sup>15</sup> Note that P5 is implied by this property.

GP3 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) \leq \sigma(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \emptyset$  and  $\mathcal{R}^+(\alpha) \neq \emptyset$  then  $\tau(\alpha) \leq \sigma(\alpha)$ .

GP3 for  $\sigma \not\perp$  and  $<$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\exists \beta \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\tau(\alpha) < \sigma(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\nexists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\exists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\tau(\alpha) < \sigma(\alpha)$ .

GP3 for  $\sigma \not\perp$  and  $<_{\perp}$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\exists \beta \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\tau(\alpha) < \sigma(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \perp$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\nexists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\exists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\tau(\alpha) < \sigma(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \perp$ .

Again, the latter two properties may be useful if arguments with bottom base score need to have special status.

GP3 for  $\sigma \not\perp$  and  $<_{\top}$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\tau(\alpha) < \sigma(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \top$ .<sup>16</sup>
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\nexists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\exists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\tau(\alpha) < \sigma(\alpha)$  or  $\tau(\alpha) = \sigma(\alpha) = \top$ .

GP3 for  $\sigma \not\perp$  and  $\leq$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF, if  $\exists \beta \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\beta) > \perp$  then  $\tau(\alpha) \leq \sigma(\alpha)$ .
- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\nexists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$  and  $\exists \gamma \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\gamma) > \perp$  then  $\tau(\alpha) \leq \sigma(\alpha)$ .

The hyphens in Table 2 in the rows for GP4 correspond to the following properties:

GP4 for  $\sigma \perp$ :

- For any  $\alpha \in \mathcal{X}$  in an aQBAF or in a QBAF, if  $\sigma(\alpha) < \tau(\alpha)$  then  $\mathcal{R}^-(\alpha) \neq \emptyset$ .

GP4 for  $\sigma \not\perp$ :

- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\sigma(\alpha) < \tau(\alpha)$  then  $\exists \beta \in \mathcal{R}^-(\alpha)$  such that  $\sigma(\beta) > \perp$ .

The hyphens in Table 2 in the rows for GP5 correspond to the following properties:

GP5 for  $\sigma \perp$ :

- For any  $\alpha \in \mathcal{X}$  in an sQBAF or in a QBAF, if  $\sigma(\alpha) > \tau(\alpha)$  then  $\mathcal{R}^+(\alpha) \neq \emptyset$ .

GP5 for  $\sigma \not\perp$ :

- For any  $\alpha \in \mathcal{X}$  in a QBAF, if  $\sigma(\alpha) > \tau(\alpha)$  then  $\exists \beta \in \mathcal{R}^+(\alpha)$  such that  $\sigma(\beta) > \perp$ .

The hyphens in Table 2 in the rows for GP7 correspond to the following properties:

GP7 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \subsetneq \mathcal{R}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$ .<sup>17</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) \subsetneq \mathcal{R}^-(\beta)$  and  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$ .

GP7 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \subsetneq \mathcal{R}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) \subsetneq \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .

<sup>16</sup> Note that P6 is implied by this property.

<sup>17</sup> Note that P17 is implied by this property.



GP7 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \subsetneq \mathcal{R}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) \subsetneq \mathcal{R}^-(\beta)$  and  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .

GP7 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \subsetneq \mathcal{R}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .<sup>18</sup>

GP7 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$ .

GP7 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .<sup>19</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .

GP7 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .

GP7 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .

The hyphens in Table 2 in the rows for GP8 correspond to the following properties:

GP8 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \subsetneq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) \subsetneq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .

GP8 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \subsetneq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) \subsetneq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .

GP8 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \subsetneq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) \subsetneq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .

GP8 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \subsetneq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .

GP8 for  $\sigma \perp$  and  $<$ :

<sup>18</sup> Note that P18 is implied by this property.

<sup>19</sup> Note that P19 and P20 are implied by this property.

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .

GP8 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .

GP8 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .<sup>20</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .

GP8 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subsetneq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .

The hyphens in Table 2 in the rows for GP9 correspond to the following properties:

GP9 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .

GP9 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .<sup>21</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .

GP9 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .<sup>22</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .

GP9 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .

GP9 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) < \tau(\beta)$  then  $\sigma(\alpha) < \sigma(\beta)$ .

GP9 for  $\sigma \perp$  and  $<_{\perp}$ :

<sup>20</sup> Note that P22 is implied by this property.

<sup>21</sup> Note that P23 is implied by this property.

<sup>22</sup> Note that P24 is implied by this property.

- <sup>23</sup> Note that P26 and P27 are implied by this property.

GP10 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \mathcal{R}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) <_{\sigma \perp} \mathcal{R}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .

The hyphens in Table 2 in the rows for GP11 correspond to the following properties:

GP11 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$ .

GP11 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .

GP11 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .<sup>24</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\beta) = \sigma(\alpha) = \top$ .

GP11 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .

GP11 for  $\sigma \perp$  and  $<$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$ .

GP11 for  $\sigma \perp$  and  $<_{\perp}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \perp$ .

GP11 for  $\sigma \perp$  and  $<_{\top}$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\alpha) = \sigma(\beta) = \top$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) < \sigma(\alpha)$  or  $\sigma(\beta) = \sigma(\alpha) = \top$ .

GP11 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) >_{\sigma \perp} \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .

To conclude, we give examples of some of the hyphens in Table 3:

GP6 and GP7 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \subseteq \mathcal{R}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .<sup>25</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) \subseteq \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .

GP6 and GP7 for  $\sigma \perp$  and  $\leq$ :

<sup>24</sup> Note that P28 and P29 are implied by this property.

<sup>25</sup> Note that P30 is implied by this property.

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subseteq \mathcal{R}_{\sigma \perp}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) \subseteq \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\beta) \leq \sigma(\alpha)$ .

GP6 and GP8 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}^+(\alpha) \subseteq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .<sup>26</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) = \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) \subseteq \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .

GP6 and GP8 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an sQBAF, if  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subseteq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}_{\sigma \perp}^-(\alpha) = \mathcal{R}_{\sigma \perp}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) \subseteq \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \leq \sigma(\beta)$ .

GP6 and GP10 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \leq_{\sigma \perp} \mathcal{R}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \geq \sigma(\beta)$ .<sup>27</sup>
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF,  $\mathcal{R}^-(\alpha) \leq_{\sigma \perp} \mathcal{R}^-(\beta)$ ,  $\mathcal{R}^+(\alpha) = \mathcal{R}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \geq \sigma(\beta)$ .

GP6 and GP10 for  $\sigma \perp$  and  $\leq$ :

- For any  $\alpha, \beta \in \mathcal{X}$  in an aQBAF, if  $\mathcal{R}^-(\alpha) \leq_{\sigma \perp} \mathcal{R}^-(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \geq \sigma(\beta)$ .
- For any  $\alpha, \beta \in \mathcal{X}$  in a QBAF, if  $\mathcal{R}^-(\alpha) \leq_{\sigma \perp} \mathcal{R}^-(\beta)$ ,  $\mathcal{R}_{\sigma \perp}^+(\alpha) = \mathcal{R}_{\sigma \perp}^+(\beta)$  and  $\tau(\alpha) = \tau(\beta)$  then  $\sigma(\alpha) \geq \sigma(\beta)$ .

## References

- [1] P.M. Dung, On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games, *Artif. Intell.* 77 (2) (1995) 321–358, [https://doi.org/10.1016/0004-3702\(94\)00041-X](https://doi.org/10.1016/0004-3702(94)00041-X).
- [2] L. Amgoud, C. Cayrol, M. Lagasque-Schie, P. Livet, On bipolarity in argumentation frameworks, *Int. J. Intell. Syst.* 23 (10) (2008) 1062–1093.
- [3] L. Amgoud, J. Ben-Naim, Evaluation of arguments from support relations: axioms and semantics, in: *Proceedings of the Twenty-Fifth International Joint Conference on Artificial Intelligence, IJCAI, 2016*, pp. 900–906.
- [4] M.W.A. Caminada, D.M. Gabbay, A logical account of formal argumentation, *Stud. Log.* 93 (2–3) (2009) 109–145, <https://doi.org/10.1007/s11225-009-9218-x>.
- [5] P. Matt, F. Toni, A game-theoretic measure of argument strength for abstract argumentation, in: *Logics in Artificial Intelligence, Eleventh European Conference, JELIA, 2008*, pp. 285–297.
- [6] D.M. Gabbay, Equational approach to argumentation networks, *Argum. Comput.* 3 (2–3) (2012) 87–142, <https://doi.org/10.1080/19462166.2012.704398>.
- [7] L. Amgoud, J. Ben-Naim, Ranking-based semantics for argumentation frameworks, in: *Scalable Uncertainty Management, Seventh International Conference, SUM, 2013*, pp. 134–147.
- [8] L. Amgoud, J. Ben-Naim, D. Doder, S. Vesic, Ranking arguments with compensation-based semantics, in: *Principles of Knowledge Representation and Reasoning: Proceedings of the Fifteenth International Conference, KR, 2016*, pp. 12–21, <http://www.aaai.org/ocs/index.php/KR/KR16/paper/view/12816>.
- [9] P. Baroni, M. Romano, F. Toni, M. Aurisicchio, G. Bertanza, Automatic evaluation of design alternatives with quantitative argumentation, *Argum. Comput.* 6 (1) (2015) 24–49, <https://doi.org/10.1080/19462166.2014.1001791>.
- [10] J. Leite, J. Martins, Social abstract argumentation, in: *Proceedings of the Twenty-Second International Joint Conference on Artificial Intelligence, IJCAI, 2011*, pp. 2287–2292, <http://ijcai.org/papers11/Papers/IJCAI11-381.pdf>.
- [11] M. Thimm, G. Kern-Isberner, On controversiality of arguments and stratified labelings, in: *Computational Models of Argument – Proceedings of COMMA 2014, 2014*, pp. 413–420.
- [12] L. Amgoud, J. Ben-Naim, Axiomatic foundations of acceptability semantics, in: *Principles of Knowledge Representation and Reasoning: Proceedings of the Fifteenth International Conference, KR, 2016*, pp. 2–11, <http://www.aaai.org/ocs/index.php/KR/KR16/paper/view/12855>.
- [13] E. Bonzon, J. Delobelle, S. Konieczny, N. Maudet, A comparative study of ranking-based semantics for abstract argumentation, in: *Proceedings of the Thirtieth Conference on Artificial Intelligence, AAAI, 2016*, pp. 914–920, <http://www.aaai.org/ocs/index.php/AAAI/AAAI16/paper/view/12465>.
- [14] L. Amgoud, J. Ben-Naim, D. Doder, S. Vesic, Acceptability semantics for weighted argumentation frameworks, in: *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence, IJCAI, 2017*, pp. 56–62.
- [15] L. Amgoud, J. Ben-Naim, Evaluation of arguments in weighted bipolar graphs, in: *Symbolic and Quantitative Approaches to Reasoning with Uncertainty – Proceedings of the Fourteenth European Conference, ECSQARU 2017, Lugano, Switzerland, July 10–14, 2017*, pp. 25–35.
- [16] P. Baroni, A. Rago, F. Toni, How many properties do we need for gradual argumentation?, in: *Proceedings of the Thirty-Second Conference on Artificial Intelligence, AAAI, New Orleans, Louisiana, USA, February 2–7, 2018*, 2018, <https://www.aaai.org/ocs/index.php/AAAI/AAAI18/paper/view/16280>.
- [17] A. Rago, F. Toni, Quantitative argumentation debates with votes for opinion polling, in: *PRIMA 2017: Principles and Practice of Multi-Agent Systems – Proceedings of the Twentieth International Conference, Nice, France, October 30–November 3, 2017*, pp. 369–385.
- [18] M. Beirlaen, J. Heyninck, P. Pardo, C. Strasser, Argument strength in formal argumentation, *IfCoLog J. Log. Appl.* 5 (3) (2018) 629–675.
- [19] W. Peden, Imprecise probability and the measurement of Keynes's "weight of arguments", *IfCoLog J. Log. Appl.* 5 (3) (2018) 677–707.
- [20] A. Rago, F. Toni, M. Aurisicchio, P. Baroni, Discontinuity-free decision support with quantitative argumentation debates, in: *Principles of Knowledge Representation and Reasoning: Proceedings of the Fifteenth International Conference, KR, 2016*, pp. 63–73, <http://www.aaai.org/ocs/index.php/KR/KR16/paper/view/12874>.
- [21] A. Rago, P. Baroni, F. Toni, On instantiating generalised properties of gradual argumentation frameworks, in: *Scalable Uncertainty Management – Proceedings of the Twelfth International Conference, SUM 2018, Milan, Italy, October 3–5, 2018*, pp. 243–259.

<sup>26</sup> Note that P31 is implied by this property.

<sup>27</sup> Note that P32 and P33 are implied by this property.

- [22] P. Baroni, G. Comini, A. Rago, F. Toni, Abstract games of argumentation strategy and game-theoretical argument strength, in: *PRIMA 2017: Principles and Practice of Multi-Agent Systems – Proceedings of the Twentieth International Conference*, Nice, France, October 30–November 3, 2017, 2017, pp. 403–419.
- [23] P. Besnard, A. Hunter, A logic-based theory of deductive arguments, *Artif. Intell.* 128 (1–2) (2001) 203–235, [https://doi.org/10.1016/S0004-3702\(01\)00071-6](https://doi.org/10.1016/S0004-3702(01)00071-6).
- [24] P. Besnard, V. David, S. Doutre, D. Longin, Subsumption and incompatibility between principles in ranking-based argumentation, in: *Twenty-Ninth IEEE International Conference on Tools with Artificial Intelligence, ICTAI 2017*, Boston, MA, USA, November 6–8, 2017, 2017, pp. 853–859, <http://doi.ieeecomputersociety.org/10.1109/ICTAI.2017.00133>.
- [25] E. Bonzon, J. Delobelle, S. Konieczny, N. Maudet, A parametrized ranking-based semantics for persuasion, in: *Scalable Uncertainty Management – Proceedings of the Eleventh International Conference, SUM 2017*, Granada, Spain, October 4–6, 2017, 2017, pp. 237–251.
- [26] D. Grossi, S. Modgil, On the graded acceptability of arguments, in: *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence, IJCAI 2015*, Buenos Aires, Argentina, July 25–31, 2015, 2015, pp. 868–874, <http://ijcai.org/Abstract/15/127>.
- [27] T. Patkos, A. Bikakis, G. Flouris, A multi-aspect evaluation framework for comments on the social web, in: *Principles of Knowledge Representation and Reasoning: Proceedings of the Fifteenth International Conference, KR 2016*, Cape Town, South Africa, April 25–29, 2016, 2016, pp. 593–596, <http://www.aaai.org/ocs/index.php/KR/KR16/paper/view/12885>.
- [28] P.E. Dunne, A. Hunter, P. McBurney, S. Parsons, M. Wooldridge, Weighted argument systems: basic definitions, algorithms, and complexity results, *Artif. Intell.* 175 (2) (2011) 457–486, <https://doi.org/10.1016/j.artint.2010.09.005>.
- [29] D. Dubois, H. Fargier, J. Bonnefon, On the qualitative comparison of decisions having positive and negative features, *J. Artif. Intell. Res.* 32 (2008) 385–417, <https://doi.org/10.1613/jair.2520>.